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«The Influence of Acquirer's Credit Rating on M&A Deals Performance in Russia»

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Abstract

The study investigates the influence of M&A deal on the probability of an acquirer's credit rating downgrade and the association between an acquirer's credit rating characteristics and the performance of M&A deals with a sample of 201 deals conducted during the period from 2000 to 2022 inclusive by Russian acquiring companies. Firstly, we test the hypothesis on how deal-related variables affect the probability of an acquirer's credit rating downgrade caused exactly by the M&A deal using probability model and applying the maximum likelihood method. The results show that cross-border deals increase the probability of an acquirer's credit rating downgrade by 6.5 percentage points, while stock payment method increases the probability of an acquirer's credit rating downgrade by 8.6 percentage points. Secondly, we test the hypothesis on the influence of an acquirer's credit rating characteristics on M&A performance applying an event-study and the ordinary least squares method. The results of the study show that acquirer's credit rating and probability of subsequent acquirer's credit rating downgrade decrease the M&A performance of acquirers in the Russian market. This study has a practical implication for investors as, firstly, it allows to predict an acquirer's credit rating downgrade as result of the deal which is associated with an increase in default risk and higher cost of debt financing. Secondly, based on a bidder's credit rating characteristics investors will be able to assess whether a certain M&A deal is value creating or a value destroying.

Keywords: Credit rating, M&A deals, Event study, Credit rating downgrade, National rating scale

1. Introduction

Credit rating plays an essential role in the capital market. It is considered as an indicator of the relative credit quality of a company for investors and a key determinant of the cost of debt financing. For these reasons, credit ratings (CRs) not only result in changes in the debt payments, but also affect the behaviour of the company. However, according to credit rating agencies' (CRAs) press releases, acquirers often put their credit ratings at risk in mergers and acquisitions (M&A) because these deals often result in CR downgrades as they are associated with large cash outflows and increased debt burden (*Chung et al., 2012*). For the company, a loss of a credit rating is associated with a significant increase in the cost of debt financing and the risk of default, which becomes especially important in periods of high interest rates in the economy. For investors, the risk of a credit rating downgrade of a company is also a negative signal and may affect stock returns. Moreover, the credit rating level determines the acquirer's financial constraints, which should be also taken into account in M&A deal evaluation. The factors mentioned above suggest that there is a relationship between CR characteristics and M&A performance.

It is a common knowledge that M&A deals do not always bring economic value to the acquirer. According to KPMG research, 83% of mergers were unsuccessful in generating any business benefit in terms of shareholder value (*KPMG, 1999*). The problems of the probability of acquirer's credit rating review as a result of the deal and the impact of acquirer's credit rating characteristics on M&A deals performance are widely discussed in the literature. In previous studies, the probability of CR downgrade after a merger have been associated with certain characteristics of the deal, acquirer and the target company (*Kiesel, Riehl, 2024; Chang et al., 2018; Chen, 2003*). There is also an evidence in the academic literature that investors can predict the actual CR revision in advance (*Kenjegaliev et al., 2016*) and that CR change risk has a significant negative influence on M&A performance, measured by the stock market reaction to a deal announcement (*Kiesel, Riehl, 2024*). Previous studies also found that there is a significant controversial relationship between the absolute level of acquirer's CR with the M&A performance (*Aktas et al., 2021; Kang, 2022; Kiesel, Riehl, 2024; Blomkvist et al., 2024; Blomkvist et al., 2021; Hasan et al., 2021*).

However, all previous research in this field were conducted in developed markets. Similar studies have not been performed for Russia. In addition, previous researchers used international rating scale in their analysis (*Kiesel, Riehl, 2024; Hasan et al., 2021; Chang et al., 2018; Chen, 2003*). The existing studies in this field have not used national rating scale for this purpose. Finally,

previous researchers have used data on whether the absolute level of CR was downgraded over a year following the deal announcement as the dependent variable for their models of probability of CR downgrade (*Kiesel, Riehl, 2024; Chang et al., 2018; Chen, 2003*). They did not identify the real reasons for CR downgrades, since the rating actions might have been unrelated to the deal.

Given the abovementioned conditions and the current macroeconomic situation in Russia, the relevance of our research becomes evident, as there is seen an increased investors' attention to the level of CR, which (among other things) determines the cost of capital of the company. In the conditions of tight monetary policy and high interest rates in the Russian economy, the advantages of a company with a higher CR are obvious. At the same time, investors are currently more sensitive to the risk of the company's CR downgrade, as they expect a significant increase in the debt burden and a higher risk of default. High interest rates also lead to extensions in credit spreads between issuers of different credit quality. In such circumstances, a CR downgrade would significantly increase the company's debt financing costs and decrease its investment attractiveness.

The *purpose* of this research is to determine the influence of acquirer's credit rating and the probability of its downgrade on the M&A performance in the Russian market.

In order to achieve this purpose, the following *research tasks* were identified:

1. Identify the factors affecting M&A performance and probability of CR downgrade and analyse the methods of their evaluation.
2. Systematize contemporary literature studying the influence of credit ratings on M&A performance and the influence of deal characteristics on the probability of acquirer's credit rating downgrade.
3. Identify research gaps in academic literature in the field of study.
4. Discover the specifics of credit rating assignment by different rating agencies on international and national scales to construct the variable of credit rating level on a national scale.
5. Determine the influence of M&A deals characteristics on the probability of acquirer's CR change in the Russian market in the period from 2000 to 2022.
6. Estimate the influence of acquirer's credit rating on M&A performance in the Russian market in the period from 2000 to 2022.
7. Estimate the influence of the probability of acquirer's credit rating change on M&A performance in the Russian market in the period from 2000 to 2022.

The *object* of the study is the M&A deals conducted in the Russian market from 2000 to 2022. The *subject* of the study is the influence of acquirer's credit rating characteristics on the market reaction to the acquisition announcement performed by Russian acquiring companies.

The scientific novelty of the work is as follows:

1. The research in field of relationship between CRs and M&A performance is conducted for the first time in the Russian market. Previous studies in this field were performed for the American and Asian markets (*Kiesel, Riehl, 2024; Chang et al., 2018; Chen, 2003*). The Russian market is unique in terms of its regulation and is less efficient than developed markets. In these terms, CRs should eliminate information asymmetry between investors and acquirers. However, the unique regulatory environment of the Russian market may have caused investors to doubt the quality of CRs assigned. Consequently, the influence of CRs on abnormal stock returns of M&A deals in Russia may be different from the results obtained in studies of other markets.
2. This study applies an improved version of the dependent variable for the probability model – the *CR Event*, which captures whether a particular M&A deal is mentioned in press releases related to a CR downgrade or a negative CR forecast. This approach was not used by previous authors, who considered the CR downgrade within a year after the deal as the dependent variable, regardless of whether the acquisition was the reason for the rating revision or not (*Kiesel, Riehl, 2024; Chang et al., 2018; Chen, 2003*).

The practical implication of this research is to provide investors with an awareness of how the acquirer's CR level and the threat of CR overview can affect the stock price after an acquisition announcement. This will allow investors to consider the company's credit quality and possible CR downgrade in assessing M&A performance and making investment decisions. The study is also useful for creditors and managers, as it allows them to assess how the company's acquisition plans will affect the acquirer's CR.

The rest of the paper is organized as follows. Section 2 describes the basic concepts of CRs used in the paper, national and historical peculiarities of CRAs' activity in Russia, presents the literature review. Section 3 develops the main hypotheses and models for assessing the probability of CR downgrade as a result of the deal and the influence of CR characteristics on M&A deals performance. Section 4 provides a description of the sample used to estimate the models and describes the results obtained from hypothesis testing. Concluding section formulates the interpretation of the obtained results, the main contribution and limitations of the study.

2. Research foundations: theoretical background, specifics of assigning credit ratings to Russian companies and systematization of existing literature

2.1. Theoretical background

I. Influence of M&A deal on probability of CR downgrade

M&A deals are one of the largest investment activities of the companies, resulting in a large cash outflow. M&A deals may affect the probability of a credit rating change of the acquirer, as they affect the key financial and operational metrics of the company, which are taken into account by CRAs:

- Increased leverage: debt-financed deals increase leverage and risk of default;
- Impact on cash flows: acquisitions can both improve (due to synergies) and deteriorate (due to integration costs) free cash flow (FCF) of business;
- Operational risks: failure to successfully integrate assets could result in reduced operational efficiency and profitability.

Since the credit rating is the function of the above parameters (i.e. leverage, FCF, operating assets' risk) and these parameters are affected by the merger and acquisition transaction, ultimately the transaction also affects the credit rating.

II. Influence of CR characteristics on M&A performance

There are several channels through which an acquirer's CR characteristics can affect the M&A deal announcement returns. They are formulated in 3 main arguments in the academic literature describing the mechanism and channels of the influence of CR characteristics on M&A performance:

Financial constraints argument (*Aktas et al., 2021; Blomkvist et al., 2021; Harford, Uysal, 2014*) supposes that acquirers with lower CRs have more financial constraints. Limited access to debt markets can lead to limited investments, that are more profitable and less costly. Therefore, such firms choose investment projects and perform M&A deals with the highest returns.

In contrast, the *free cash flow argument* (*Blomkvist et al., 2021; Harford, Uysal, 2014*) suggests that higher-rated firms are expected to have fewer financial constraints, which provides them with more opportunities to realise their investment plans. However, managers of such firms can deviate from the objective of growing the wealth of their shareholders and use additional capital

opportunities, or in other words, raise debt, to make overinvestment, which is likely to result in a value decline. Consequently, CRs should be negatively related to acquirer announcement returns.

Overall, both the financial constraints and the free cash flow arguments predict a significant influence of access to debt markets on a firm's ability to make investments, perform M&A deals and on the subsequent outcomes of those investments. In accordance with financial constraints and free cash flow arguments, higher CR levels reduce capital constraints for companies and facilitates acquisitions but declines M&A performance (*Blomkvist et al., 2021*). This is *inconsistent* with the assumption that highly rated companies have greater ability to fundraise, lower cost of debt and a more robust risk profile, which should reduce their cost of capital to deal financing and reduce the risks of a new acquisition, resulting in higher deal abnormal returns.

Managing for ratings argument (*Aktas et al., 2021*) states that managers tend to maintain company's CR, so they choose the target more carefully, which increases the efficiency of M&A deals. Thus, the managers' efforts to reduce the risk of a credit rating downgrade cause the company to engage in higher performance M&A deals. According to the managing for ratings argument, the higher risk of a CR downgrade as a result of a deal reduces the M&A performance.

2.2. Main concepts of credit ratings and features of credit rating agencies' work in the Russian market

A credit rating is a credit risk measure and a relative positioning of the rated entity in terms of the probability of default (*S&P's Rating Services, 2014; Dyachkova, 2018*). CR is a key factor affecting the cost of debt and therefore the cost of capital of the company (*Grishunin, Egorova, 2022*). In Russia, CRs above BBB+ are considered investment grade ratings. For the global market, a CR above BBB– on the "Big Three" international scale is considered an investment grade rating (*S&P's Rating Services, 2014*). The distinction between investment grade and speculative grade ratings is important because many large organizations and pension funds are required to invest only in investment grade instruments (*Ng, Ariff, 2019*). Credit ratings can be assigned to financial instruments, companies or countries (i.e. sovereign CR). Hereinafter in the paper, when using a term *Credit Rating* we will refer to CRs assigned to companies, unless otherwise stated.

The assignment of a specific CR level is accompanied by a procedure of evaluating the company against all other companies rated by the CRA (*Dyachkova, 2018*). A distinguishing feature of CRs is the conversion of a large number of company's characteristics and indicators into

one single relevant metric (*Dyachkova, 2018*). Credit ratings are determined based on historical and current information, as well as expected future events and therefore have predictive power (*Avrutskaya, Maricheva, 2021; S&P's Rating Services, 2014*). CRAs reduce the information asymmetry in the capital markets by holding insider information from issuers that is not available to the general public, particularly investors (*Ng, Ariff, 2019)*. Based on a comprehensive analysis of a large amount of financial and non-financial information, the CRA provides its opinion on the firm's financial stability, reliability and creditworthiness (*Chen, Shih, 2006; Dyachkova, 2018*).

CRAs also assign one of the following forecasts to the current ratings: negative, positive, stable or developing. The forecast refers to the CRA's view on the possible change in the rating level over the next 12 months.

Rating values in many countries are used as benchmarks for most securities and public companies in various industries (*Dyachkova, 2018*). Behr and Guettler concluded that investors cannot adequately assess the creditworthiness of companies on their own, and their expectations are often overestimated (*Behr, Guettler, 2008*). Therefore, investors use CR to assess credit risks and compare the reliability of alternative investment opportunities (*S&P's Rating Services, 2014; Dyachkova, 2018*). Credit ratings can be considered as a bridge between borrowers (issuers) and lenders (investors), which reduces information asymmetry in the financial system by providing an independent view of creditworthiness (*Hertikasari, Artha, 2022; Adelson, 2012*). CRs affect securities markets in many ways, including an issuer's access to capital and the structure of deals (*Boot et al., 2005*). A high CR signals to investors that a company can raise lower-cost debt financing, has a lower cost of capital, and, therefore, can use more investment opportunities (*Grishunin, Egorova, 2022*).

The foregoing highlights the significance of a company's credit rating for *equity investors*. Academic literature provides an evidence of a relationship between CR announcements and share price performance, i.e. stock returns (*Balcilar et al., 2021*). The academic literature has for many years made efforts to find a relationship between excess stock returns and a company's credit rating. Theoretically, investors expect higher returns from higher risk investments. Investing in a firm with a low credit rating is considered risky and should therefore generate higher excess returns for the investor, hence the saying "the higher the risk, the higher the return". However, some empirical studies demonstrate the opposite results (*Lee et al., 2022*). Since theory and empirical

evidence contradict each other, many refer to this discrepancy as the financial distress anomaly or the financial distress risk puzzle (*Lee et al., 2022*).

Investors tend to seek CR stability to reduce the frequency and cost of adjustments in their portfolios (*Ng, Ariff, 2019*). As a rule, CRAs monitor events that may affect the level of credit risk of the company that has a CR from that agency. The main purpose of CRs' monitoring is to update the rating by detecting factors that may cause the CR upgrade or downgrade. The frequency of credit monitoring is determined by factors specific to a company or a group of rated organizations (*S&P's Rating Services, 2014*). The reasons for a change in credit rating are varied. Some of them may be related to overall changes in the economy and business environment, while others are more specific and are determined by circumstances affecting a particular industry or company. A CRA may change the CR or revise its forecast by changing its view on the relative level of credit risk when it identifies factors that could affect the company's creditworthiness. As long as a CR provides important information to investors, a change in the CR should result in a change in the market value of the issuer's securities. This applies to both debt and equity securities.

Confirming the above, previous studies have identified a significant negative stock market reaction to a CR downgrade in developed (*Reddy et al., 2019; Salvador, 2017; Hu et al., 2016; Behr, Guettler, 2008*) and developing markets (*Dawar et al., 2021; Ng, Ariff, 2019; Ryu et al., 2019*) using event study methodology. Avrutskaya and Maricheva obtained similar results in the Russian market (*Avrutskaya, Maricheva, 2021*). Kenjegaliev et al. found that the German stock market adjusts stock prices in advance of actual CR changes, indicating that CR revision announcements are expected by investors in advance (*Kenjegaliev et al., 2016*).

However, there were several cases in history when the quality of assigned CRs was reduced and doubted by investors. The global financial crisis of 2008 challenged the credibility of CRAs as they failed to properly assess and disclose the inherent risks of the assets, causing the financial system to be on the brink of collapse (*Balcilar et al., 2021*). Previous studies also showed that increased competition in the rating industry is associated with lower CR quality: CR levels go up, the correlation between ratings and market-implied yields declines, and the ability of ratings to predict default deteriorates (*Jiang, Packer, 2019; Becker, Milbourn, 2011*). The main reasons for incorrect CR assignment are:

- Conflicts of interest (companies pay CRAs for CRs assigned) (*Financial stability forum, 2009; International Monetary and Financial Committee, 2008; Dyachkova, 2018; Balcilar*

et al., 2021);

- Shortcomings in the rating methodology (*Financial stability forum, 2009; International Monetary and Financial Committee, 2008; Dyachkova, 2018; Balcilar et al., 2021*);
- Failure to respond in a timely manner to changes in the issuer's financial position (*Dyachkova, 2018*).

Rating scale is a system of CRA's assessment of a company's ability to fulfil its financial obligations. There are several classifications of CRs' scales, we will only describe those that were used in the paper. Depending on the covered period of time, there are short-term and long-term CRs. The rating scales are also divided into national and international ones. On a national scale, CRA evaluates the issuer of securities relative to companies or financial obligations in that country. Conversely, the international scale rating allows comparing issuers across the world. The rating of an issuer on the international scale depends on the sovereign rating (i.e. the risks of doing business in a given country are taken into account) and cannot exceed it (*Dyachkova, 2018*). The CRAs publish on their official websites the methodology of converting credit ratings applied on international scale to credit ratings applied on national scales (*ACRA, 2024; Expert RA, 2025*). In this paper, long-term credit ratings on a national scale are used for analysis, which allows us to eliminate country risk and focus specifically on the risks associated with the company's business and industry. The choice of the national scale is also based on the specifics of our sample, which is represented only by Russian acquirers and a share of cross-border transactions of 14%.

The rating industry is represented by both international and national rating agencies, among which the standards for assigning CRs may vary significantly (*Dyachkova, 2018; Karminsky, Dyachkova, 2016*). The "Big Three" of foreign credit rating agencies – Moody's, Standard & Poor's, and Fitch – operate in the international credit rating services market. The "Big Three" evaluate issuers, in fact, from the investors' side and take into account the assigned sovereign ratings in international CR scale. At the same time, companies that do not plan to enter foreign markets need rating agencies familiar with local practices and business culture, so they often resort to the services of national CRAs (*Dyachkova, 2018*). As of 2025, there are 4 national credit rating agencies accredited and registered by the Bank of Russia that are operating in the Russian market: RAEX (Expert RA), ACRA (Analytical Credit Rating Agency), NCR (National Credit Ratings) and NRA (National Rating Agency).

Due to the lack of a generally accepted rating methodology, the assessment of issuers' creditworthiness becomes more complicated. Rating agencies often assign different credit ratings to the same company. A unified rating space is being developed in Russia to address this issue. The Bank of Russia publishes Comparison of rating scales of Russian and the "Big Three" credit rating agencies (*Bank of Russia, 2024; Bank of Russia, 2021; CBonds, 2014*). Tables of rating correspondence from international agencies are published by the Bank for International Settlements (*Bank for International Settlements, 2014*), Bank of Russia (*Bank of Russia, 2021*), the Ministry of Finance of Russia (*Ministry of Finance of Russia, 2014*), credit rating agencies (*Expert RA, 2025*) and CBonds (*CBonds, 2024; CBonds, 2014*). There are also a number of studies devoted to the development of tables comparing credit ratings assigned by national and international rating agencies on international and national rating scales (*Aivazian et al., 2011; Golovan, Karminsky, Peresetsky, 2011; Karminsky et al., 2013; Karminsky, Dyachkova, 2016; Dyachkova, 2018*).

In 2007, the market was roughly equally divided between Russian and foreign CRAs of the "Big Three", whose share was ensured by leadership in the banking sector, while the growth of national CRAs was mainly due to active development in the corporate sector, where they had sustainable competitive advantages (*Dyachkova, 2018*). In subsequent years, the share of national CRAs in the Russian CR services market has steadily increased.

On 14 July 2017, the Federal Law “On the Activities of Credit Rating Agencies in the Russian Federation” came into force. This law prohibited international rating agencies from assigning CRs on a national scale to Russian companies. Therefore, S&P, Moody's and Fitch have assigned only international scale credit ratings to Russian companies since 2017. In 2022, the EU prohibited international rating agencies from assigning any ratings and providing any rating services to Russian Federation and Russian companies. Thus, only national CRAs currently operate in the Russian market, which are isolated from the influence of the "Big Three" international CRAs. National CRAs are under increased pressure at this time: the confidence of investors and other financial market participants depends on the quality of credit ratings assigned.

In terms of macroeconomic conditions, the importance of CRs for investors increases during periods of tight monetary policy due to high interest rates in the economy and widening of spreads between different CR groups. In such periods, the company’s debt burden and default risk increases significantly. A downgrade in credit rating means a significant increase in debt obligations and possible deterioration of creditworthiness and sometimes even bankruptcy. Our

research sample covers several periods of tight monetary policy and significant key rate increases caused by crises and geopolitical instability: the global crisis in 2008-2009, geopolitical pressure and sanctions in 2014 and 2022. The study is especially relevant in the current conditions, as since 2023, the Bank of Russia has followed a tight monetary policy. High interest rates led to an increase in credit spreads between issuers of different credit quality at the end of Q4 2024. As of December 2024, the spread between 'A' and 'BBB' ratings was 5.8% (*Bank of Russia, 2025*). Consequently, considering all of the above mentioned, it is assumed that investors should be most sensitive to the credit rating and the probability of credit rating downgrade of Russian companies at this moment.

2.3. Impact of M&A deals on the probability of acquirer's credit rating revision: a review of previous studies

After studying 150 deals in the US with deal value of $5 billion or more since 2000, S&P found that in 25% of cases, the acquirer's initial CR was downgraded by one or more notches after the deal was announced (*Ryan, 2013*). In accordance with S&P's analysis, in the long run 53% of buyers eventually receive a CR downgrade (*Ryan, 2013*).

Chung et al. found that mergers and acquisitions announcements of the rated firms are one of the most frequent reasons for formal revisions of firms' credit ratings, but only half of them result in actual credit rating adjustments (*Chung et al., 2012*). For proposed acquisitions, CRAs often announce the credit rating review on the *same day* as the acquisition announcement or during the subsequent days, indicating that CRA analysts monitor the company's business activity in a timely and consistent manner (*Kiesel, Riehl, 2024*). By putting a rating on review for downgrade, CRAs can influence the risk decisions of companies, including M&A deals, by signaling a possible downgrade if the company fails to mitigate the risk within a certain period of time. This implicit contract between the CRA and the company implies that failure to eliminate risks may result in a downgrade, leading to negative investor reaction and increased refinancing costs (*Kiesel, Riehl, 2024*).

According to S&P, the following risks lead to CR downgrades as a result of the deal: weaker pro-forma credit terms, lower free operating cash flow and concerns about the business risk profile of the combined company (*Ryan, 2013*). Academic literature highlighted the following reasons for the increased probability of acquirer CR downgrade: higher financial leverage after a merger as a result of debt raising for the deal financing, aggressive managerial actions explained by personal

benefits for managers from the deal, transfer of risk from the target, lower credit quality of a combined firm as a result of the deal (*Kiesel, Riehl, 2024; Furfine, Rosen, 2011*).

With the expectation of continued strong corporate bond market and extremely low interest rates, more companies will put their CRs at risk by financing large M&As with large amounts of debt relative to equity and cash flow. In times of a stable economy, the difference in spreads between CRs is quite small, so companies are willing to finance an acquisition with higher debt burden. According to S&P, in the period of steady economic growth in the US, loose monetary policy and soft financing conditions during the period of 2010–2019, the downgrades of CRs to 'BBB' level in 45% of times were caused by M&A-related activities (*Kraemer, Gunter, 2019*). All issuers whose CRs were downgraded following M&A-related decisions were able to maintain investment grade CRs, mainly due to the subsequent reduction in leverage.

There are many examples in the international and Russian practice when a relatively large M&A deal (i.e. its announcement) caused a revision of the CR of a company by a rating agency:

1) In 2006, S&P placed Pyaterochka's BB international scale CR and ruAA national scale CR on CreditWatch with a negative forecast after the announcement of acquisition of Perekrestok supermarket chain for $1.6 billion, of which $300 million was to be paid in cash and $1.3 billion in stock (*CBonds, 2006*). The CreditWatch rating reflects the S&P's concern about Pyatyorochka's increased consolidated financial risk following the deal. The CRA also expected the ND/EBITDA ratio to rise to 3.0x from 2.0x-2.5x previously.
2) S&P downgraded Verizon's CR to BBB+ from A as a result of Verizon Communications acquisition of 45% of Vodafone Wireless for $130 billion (*Kraemer, Gunter, 2019*). S&P expected Verizon's leverage to increase from 3.0x to 3.4x and forecasted Verizon's "funds from operations to debt" to fall to the low 20% range.
3) Following Thermo Fisher Scientific's announcement of its $17 billion acquisition of Life Technologies, S&P downgraded the company's CR to BBB from A– (*Ryan, 2013*). It also assigned a CreditWatch rating of "CreditWatch with negative forecast". Thermo Scientific did not disclose details on the deal financing, but said it will use $9 billion in debt, increasing the leverage to about 4.5x.

The interest of researchers in the field of CR downgrade risk in the context of mergers and acquisitions is explained by the two unique characteristics of M&A deals (*Kang, 2022*). Firstly, deals have a huge value compared to the value of other types of investments. Therefore, post-

merger outcomes will have a significant impact on the default risk of the acquiring party and hence on its CR level. In addition, acquisitions may result in changes to the capital structure as they typically require the acquiring party to issue debt and/or equity to finance merger costs. Such changes in the capital structure may reduce the acquiring party's cash balance and increase its level of debt, thereby increasing the risk of the company's CR downgrade.

In accordance with *managing for ratings argument*, firms aim to maintain CRs above certain levels in the context of M&A deals (*Kang, 2022; Aktas et al., 2021; Blomkvist et al., 2021*). Previous studies have found that managing for ratings argument is more relevant for cash-financed M&A deals than for stock-financed ones (*Aktas et al., 2021*). Stock-financed mergers are less likely to put downward pressure on CRs. This can be explained by the fact that most cash deals involve raising debt, which emphasises the importance of access to the debt capital market for deal financing provided by CRs (*Harford, Uysal, 2014*).

The earliest researches stated that corporate mergers should increase the diversification effect and result in a risk mitigation of the merged company (*Amihud, Lev, 1981*). At the same time, a lot of studies showed that M&A deals increase acquirer's default risk (*Furfine, Rosen, 2011; Billett et al., 2004*). The systematization of existing research is presented in Table 1, Appendix 1.

The obtained results of the previous studies showed a significant influence of corporate acquisitions on future acquirer's CR downgrade. In particular, larger M&A deals are more likely to be associated with following CR downgrades (*Kiesel, Riehl, 2024; Furfine, Rosen, 2011*). Conversely, Chang et al. provided an evidence that mergers with lower relative deal value increase the probability of an acquirer's CR upgrade over the next 3 to 5 years after the deal announcement and simultaneously decrease the probability of a CR downgrade (*Chang et al., 2018*).

At the same time, contradictory results on the effect of payment method were observed: some authors demonstrated that stock-financed deals significantly reduce the risk of CR downgrade (*Furfine, Rosen, 2011*), while other studies claim that the payment method has no effect on the CR review after the deal announcement (*Kiesel, Riehl, 2024*). Chang et al. showed that stock-financed deals increase the probability of CR upgrade and decrease the probability of CR downgrade (*Chang et al., 2018*).

Previous papers also did not find a significant relationship between cross-border deals and the probability of CR downgrade (*Kiesel, Riehl, 2024*). There was only one study that found an

insignificant influence of M&A deals on acquirer's CR change (*Kang, 2022*). However, Kang also found that companies facing CR downgrade risk tend to perform acquisitions that can reduce the risk of future CR downgrades.

While previous literature was focused on the relationship between CRs and merger activity, there is limited number of studies on the effect of certain M&A deal characteristics on the probability of an acquirer's CR downgrade, which were conducted for the American and the Asian markets only (*Kiesel, Riehl, 2024; Chang et al., 2018; Chen, 2003*). There were no relevant studies for the Russian market. The Russian market is less efficient than developed ones and therefore the performance of CRAs aimed at eliminating information asymmetry between issuers (in the acquisition process) and investors, as well as their reaction to M&A deal announcements, may differ from those agencies of the developed markets. Moreover, as mentioned above, since 2017 restrictions were imposed on the activities of the "Big Three" international CRAs in relation to Russian companies, hence the CRAs' activity on the Russian market is interesting within this topic due to its isolation.

Previous researchers have used data on whether the absolute level of CR was downgraded over a year following the deal announcement as the dependent variable for their models of probability of CR downgrade (*Kiesel, Riehl, 2024; Chang et al., 2018; Chen, 2003*). The main drawback of such approach is that it does not take into account the information provided by the credit rating agencies press releases and therefore does not identify the real reasons for credit rating downgrades, since the rating actions might have been unrelated to the deal (e.g. country CR downgrade (for the international scale), parent company downgrade, industry performance decline, corporate governance deterioration and etc.). Thus, it is vitally important to bridge this gap in order to study the real cause, whether a certain M&A deal announcement led to a CR downgrade or not.

Summing up the literature studied in this paragraph, we identified two research gaps, which concern the lack of research on CRs revisions as a result of the deal in the Russian market, as well as the lack of literature which utilizes the approach based on the *ex-ante* analysis of data from real press releases that mention CR downgrades as a result of a specific deal, rather than any other reason for CR downgrade. In this study we are going to explicitly connect the CR revision with the acquisition announcement as the reason for such downgrade.

2.4. Influence of acquirer's credit rating characteristics on M&A performance: a new angle

Previous researches provided an evidence that numerous determinants can influence acquirer's M&A deal announcement returns. Specific features of abnormal returns analysis in the Russian market are: a) illiquidity of stock market (*Bertrand, Betschinger, 2012*), b) insignificant or small influence of M&A deals on shareholder's value (Appendix 1, Table 2), and c) event-study analysis in the Russian market is conducted on a relatively small sample as equity capital market is less developed than mature markets (usually the period of observation starts from year of 2000, while for developed markets it could be even from the beginning of 1980's) and there is a limited number of Russian public companies. A sample of 150-200 observations of M&A deals on the Russian market is considered to be representative (Appendix 1, Table 2).

Further on, we will focus on the influence of an acquirer's CR characteristics on M&A performance (Appendix 1, Table 3).

Influence of an assigned acquirer's CR on M&A performance

Aktas et al. found a curvilinear relationship between CR and M&A stock announcement returns (*Aktas et al., 2021*). For the acquirers with non-investment grade CRs, corporate acquisition returns initially decline with CR growth as it expands the firm's ability to engage in fundraising and overinvestment in takeovers. This confirms the *financial constraints argument*. In the upper echelons of CR range, the returns associated with M&A deal announcements increase along with the CR increase. This is consistent with the *managing for ratings argument*, since if such companies are trying to maintain investment grade CR, they should only undertake the most profitable acquisitions and choose only the most attractive targets, refusing other deals unless they are particularly interested in takeovers.

However, there is also a number of studies that found no significant relationship between CR level and M&A performance (*Kiesel, Riehl, 2024; Hasan et al., 2021; Blomkvist et al., 2021*).

Prior studies found a significant influence on M&A performance of not a certain level of acquirer's CR, but the fact that the acquirer has a CR. Harford and Uysal showed that equity capital markets react more negatively to acquisition announcements initiated by credit-rated acquirers, as it was found that credit-rated companies pay a higher premium compared to ones that do not have a CR (*Harford, Uysal, 2014*). This result confirms the *financial constraints argument*, as the CR presence and access to debt capital markets has a negative effect on M&A performance, which may

be related to the quality of targets. In particular, managers of non-credit-rated firms tend to pursue acquisitions that create higher value for shareholders compared to managers of rated firms, whose reputation and public image are already proved on the market.

Blomkvist et al. provided an evidence of a positive influence of initial CR assignment on deal announcement abnormal returns (*Blomkvist et al., 2024; Blomkvist et al., 2021*). This confirms the value creation hypothesis: companies tend to access the debt capital market in order to create shareholder value through corporate acquisitions.

Influence of probability of an acquirer's CR downgrade on M&A performance

Academic literature connected the acquisition announcement (the triggering event) with the subsequent negative acquirer's CR event (the CRA response): CR downgrade or placing the CR under review for downgrade. There are a number of studies examining the impact of the probability of a subsequent negative CR event on the stock market reaction to the deal announcement.

Kiesel and Riehl provided an evidence that the stock market considers the probability of an acquirer's CR downgrade as deals with the increased risk of CR downgrade show lower announcement abnormal returns (*Kiesel, Riehl, 2024*). The CR downgrade risk variable was calculated as the within-sample fitted probability, estimated from the probit model. The authors also demonstrated the lack of significant stock market response to the CRAs' announcement of a CR review associated with the acquisition announcement. The authors explained this observation by two factors: CRA's quick reaction and the overlap with the M&A deal announcement. The result concludes that investors can predict the probability of a CR downgrade in advance and that the announcement of a CR review for downgrade after an acquisition announcement is not a new information for the stock market.

There were also other approaches to measure an acquirer's CR downgrade risk after a deal announcement. Kang argued that if a company's broad CR has a minus sign (the so-called "minus firms"), it means that there is a probability of a CR downgrade to the next rating category (e.g. if a firm is rated "A–", it means that the firm's rating group may be downgraded from "A" to "BBB") (*Kang, 2022*). The results showed that there is no significant relationship between CR downgrade risk and M&A deal abnormal returns.

We also consider that the existence of disagreements in a company's CRs assigned by different CRAs (the so-called "split ratings") is an indirect indicator of the risk of future CR

revisions, in particular CR downgrades. If a particular CRA assigned a lower CR, this may indicate detected weaknesses and inherent risks that other CRAs have not yet taken into account. Also assigning different CRs can cause investors to doubt their quality, so CRAs try to avoid this. Hasan et al. showed that split CRs affect negatively the M&A performance, which indirectly indicates the negative impact of CR downgrade risk on deal announcement stock market reaction (*Hasan et al., 2021*).

New angle on the influence of CR characteristics on M&A performance

The literature review revealed a large number of studies on the influence of CR on the deal process in developed and emerging markets: choice of payment method, time to complete the deal, probability of deal completion, etc. However, studies on the impact of CR on M&A performance were only conducted for developed markets (Appendix 1, Table 3). As we mentioned above, the Russian CRA market is unique due to the specifics of its regulation. Also, the Russian stock market is less efficient and less liquid than developed markets in the US and Europe. Given these peculiarities, the results of such a study in the Russian market may differ from previous researches. Thus, we conclude that the lack of research in this field in the Russian market is a *research gap* and the *novelty* of our study consists in the fact that no one has previously conducted such an analysis for emerging markets.

There is only one paper (*Kiesel, Riehl, 2024*) in the US market that examines the influence of the estimated probability of a *CR downgrade* as a result of the deal on abnormal deal returns. This approach assumes that investors are able to predict a future CR change (future event) based on available data at the time of the M&A deal announcement (trigger event). The authors also studied the influence of the *CR review for downgrade* variable as a result of the deal. CR review for downgrade is an *ex-post* variable because it takes some time for the CRA to issue a press release and therefore this variable is outside the time horizon of the event study. This means that this CR review information is not known to investors at the time of the deal announcement and therefore is not implied in the abnormal return. Consequently, the Kiesel and Riehl results showed that this variable is not significant (*Kiesel, Riehl, 2024*). It might be possible to obtain different results for this variable calculated using a probit model similar to the CR downgrade variable. In this case, all the information for estimating the probability of CR review for downgrade would be known to investors in advance at the time of M&A deal announcement and would not extend beyond the time horizon of the event study. In this study we construct a new variable accounting for negative

CR events in a wide sense (Section 3.2) and estimate the probability based on the information available to investors at the time of the deal announcement.

Moreover, as we mentioned earlier, the authors looked at the fact of rating changes (*CR downgrade*) over a certain period after the deal (1 year, 3 years, 5 years) without determining the specific reasons for the rating change, which may be unrelated to the deal. The absence of deep analysis to identify the reasons for the post-merger rating change in this field is a *research gap*. The *novelty* of our study is that we examined data from more than 200 press releases on companies' CR changes to identify in which cases M&A deal was the real reason for subsequent CR change (Section 3.2).

3. Hypotheses and model for assessing the probability of credit rating review as a result of a deal and the impact of acquirer's credit rating characteristics on M&A performance

3.1. Research hypotheses

To efficiently capture the full effect of credit ratings' influence on M&A deals performance we first need to remind us of the channels of transmission of credit rating impact on acquisition announcement. By the *full* effect, as discussed in the *Theoretical background*, we mean not only the absolute level of acquirer's credit rating (e.g. 'AAA' or triple-A), but also the probability of change of such rating as a result of the deal. For an investor reacting on an M&A announcement, it is vital not only the credit quality of the issuer (i.e. its credit rating), but also the probability of its loss after the M&A deal is announced and completed. So, first we will propose what determines the probability of acquirer's credit rating change as a result of the acquisition (the first set of hypotheses) and after that we address the complete impact of credit rating characteristics on M&A deals performance (the second set of hypotheses).

Based on the previous studies, we distinguish two categories of hypotheses: H1.1–H1.3 concern the impact of M&A deals characteristics on probability of acquirer's CR downgrade, H2–H3 concern the influence of acquirer's CR characteristics on M&A performance.

Hypotheses on the influence of M&A deals characteristics on probability of acquirer's CR downgrade

According to data from press releases and a review of academic literature, one of the most common reasons for a rating revision as a result of the deal is an increase in company's debt burden, higher financial leverage and deterioration of debt covenants (Net Debt/EBITDA, Interest coverage ratio and etc.). This applies primarily to debt-financed corporate acquisitions. In increases the default risk and results in a credit rating revision. Other reasons include declining free cash flow and increased risk of the operating assets of business as a result of a high deal associated costs, synergies overvaluation and business integration issues.

Large size of M&A deal implies a large bidder's cash outflow. Relatively large acquisitions generally are not capable of being 100% financed by the company's own cash on its balance sheet. Therefore, they tend to be financed by either equity or debt. According to S&P research, large deals are often debt-financed (*Ryan, 2013*). Raising debt by an acquirer increases its financial leverage and risk of default. Additionally, if the deal is financed by a significant

share of the acquiring party's own cash balance, it implies a cash balance reduction which also negatively affects debt covenants (ND/EBITDA). So, in either of the result, a large cash outflow caused by a large size of the M&A deal a significant deteriorating factor for the acquirer's financial position. Moreover, if it is an acquisition of a large business, there are significant asset integration risks, which negatively affects the bidder's risk profile and hence increase CR downgrade risk. A literature review in the USA market showed that deal value is a main factor contributing to the acquirer's CR downgrade (*Kiesel, Riehl, 2024; Furfine, Rosen, 2011*). Given the abovementioned, larger M&A deals increase the acquirer's default risk and we assume that:

H1.1: *Relative size of M&A deals has a positive influence on probability of credit rating downgrade for Russian acquirers*

The commonly used M&A deal payment methods are either cash or stock, and their combinations. Cash payment assumes that the deal can be financed either by own cash balance or with debt. Harford and Uysal argue that most cash payment deals are debt financing (*Harford, Uysal, 2014*). At the same time, debt attraction increases the financial leverage of the acquirer and the risk of credit rating downgrade, while financing and payment by stocks (outstanding or newly issued shares) has no negative impact on the company's debt burden and the risk of credit rating downgrade. Financially distressed firms can still use their stocks as a payment method even if they cannot raise additional debt capital, and such equity-financed acquisitions are less likely to put pressure on the credit rating. Chang et al. found that the stock payment method significantly reduces the risk of credit rating downgrade (*Chang et al.,2018*). Summarising the above, the stock payment method reduces the risk of bidder's credit rating downgrade.

H1.2: *Stock payment method of M&A deals has a negative influence on probability of credit rating downgrade for Russian acquirers*

The credit rating is assigned based on the company's forecasted default risk, assessed on the basis of the data available at the time of CR assignment. The credit rating may be revised in the presence of significant events that may affect the default risk. These significant events include a cross-border M&A deal as the acquirer's current rating reflects its business profile available to the agency at the time of the rating assignment. When a company enters a new foreign market, the bidder faces additional risk exposure, which was not taken into account in the assessment of

the actual rating. Performing cross-border deal, the acquirer is under the influence of new risks: country risk (in particular, political and regulatory risks), currency risk and etc. Moreover, cross-border deals increase significantly the risk of new assets integrating due to cultural differences. Taking this into account, it is possible that a cross-border deal increases the probability of a CR downgrade. Although Kiesel and Riehl study did not find a significant impact of cross-border deals on the probability of an acquirer's credit rating downgrade in the USA market (*Kiesel, Riehl, 2024*), we consider that for a Russian acquirer cross-border deal is more risky due to differences in currency risks (volatility of the Russian ruble, while the US dollar is the reserve currency), geopolitical risks and differences in managers' expertise in similar deals. Therefore, cross-border acquisitions will put downward pressure on the risk of a CR change in the Russian market.

H1.3: *Cross-border M&A deals have a positive influence on probability of credit rating downgrade for Russian acquirers*

Hypotheses on the influence of acquirer's credit rating characteristics on M&A deals performance

Now we proceed to the hypotheses concerning the influence of acquirer's credit rating characteristics on M&A performance. We define credit rating characteristics as both the absolute credit rating level and the probability of an acquirer's CR downgrade as a result of the deal.

Consistent with the financial constraints (*Aktas et al., 2021; Blomkvist et al., 2021; Harford, Uysal, 2014*) and free cash flow (*Blomkvist et al., 2021; Harford, Uysal, 2014*) arguments, as CRs increase, the financial constraints become weaker and acquirers have more overinvestment opportunities and the overall quality of corporate acquisitions decline. In contrast, bidders with lower CR level have limited access to debt capital markets, have less investment opportunities and therefore more carefully choose targets and perform M&A deals, yielding higher returns. Aktas et al. found a negative influence of CR on M&A performance for the non-investment grade credit ratings (*Aktas et al., 2021*). According to the financial constraints and free cash flow arguments and the obtained in the previous research results, we expect that credit rating reduces the M&A performance.

H2: *Acquirer's credit rating has a negative influence on the performance of M&A deals for Russian acquirers*

According to managing for ratings argument, companies are making efforts to avoid credit rating downgrade (*Aktas et al., 2021*). As a result, such acquirers tend to approach more careful target choice and perform M&A deals with higher performance. On the other hand, investors are able to predict the credit rating downgrade risk resulting from the deal, which would be negatively reflected in stock returns, as it would indicate a deterioration of the company's risk profile, increased probability of default and additional debt financing costs. Previous studies found that an acquirer's credit rating revision risk reduces M&A deal associated abnormal stock returns (*Kiesel, Riehl, 2024*). Given the abovementioned, probability of acquirer's credit rating downgrade is assumed to be negatively affecting the M&A deals performance.

H3: *Probability of acquirer's credit rating downgrade has a negative influence on the performance of M&A deals for Russian acquirers*

3.2. Research models and framework

Empirical research in this paper is carried out in two steps. During the first step, the probability of change of the acquirer's credit rating as a result of the deal is tested and investors' expectations are determined. In the second stage, the influence of acquirer's credit rating on M&A deals performance is assessed. This paragraph describes both the probit model used in the first step and the regression model with the event study used for the second step.

3.2.1. Probability of credit rating change (binary choice model)

So, first we will determine how to measure the expectations of investors about the change of the credit rating in acquisition in order to subsequently employ them in the second step of analysis. To define the *expectations* of investors about the probability of change of the credit rating, we first need to elaborate on what is the change of the credit rating in connection with a certain M&A deal by itself. Following the work of (*Kiesel, Riehl, 2024*), where a similar concept was developed, we will introduce the following variable:

Credit Rating Event (CR Event) – binary variable that takes the value of «1» if a negative press release about the acquirer's credit rating was issued due to the announcement of the M&A deal, 0 otherwise. So, if a certain credit rating agency issued a negative statement about the acquirer regarding the potential acquisition, the following deal receives a «1» in *CR Event*. This approach is more profound and beneficial to the approach of (*Kiesel, Riehl, 2024*), where the

authors simply gathered the data on post-M&A credit ratings' downgrades in one year following the acquisition. Such downgrades may not have been exactly done as a result of the deal. So, contrary to the work of (*Kiesel, Riehl, 2024*), we have explicitly and accurately analyzed each of the credit rating agencies' press releases and have made a final conclusion about the change of the acquirer's credit rating *only if* a direct mentioning of the deal was made in the press release. See examples of such press releases in the Appendix 2 (*Exhibit 1*). An additional comment has to be made in that we have accounted not only for the credit rating downgrade as the ultimate negative reaction of the credit rating agency, but also for other negatively associated rating actions, such as putting the acquirer into the credit rating watch list, change of the rating's outlook or placing the ratings on review for possible downgrade. Such rating actions have been classified as having the same nature to the credit rating downgrade, and thus the respective deals also received «1». That is, we have named this variable the *Credit Rating Event*, not the Credit Rating downgrade, since it is the reflection of a negative credit rating actions in a wider sense.

The variable introduced in the previous section is a measure of probability of acquirer's credit rating change *ex-post*, since it is the actual data gathered from a real press releases and announcements observed in the market. To make an *ex-ante* estimate of this probability, or in other words, to get the *expectations* of investors about the change of the credit rating, we need to construct a binary choice model describing this probability, where the *CR Event* will act as a dependent variable. The estimated probability from this model will yield the investors' *expectations*, which will be eventually used as the determining factor of the M&A performance.

To measure the probability of change of the credit rating of the acquirer due to the M&A announcement, we will develop the following *Credit Rating Event* binary choice model:

$$\mathbb{P}\,(\textbf{CR Event = 1}) = \mathbb{F}\,(\beta_0 + \beta_1\, \textit{Relative deal size} + \beta_2\, \textit{Payment method} + \beta_3\, \textit{Cross-border} + \\ + \beta_4\, \Delta \textit{Key rate} + \beta_5\, \textit{Public Target} + \beta_6\, \textit{ln (Revenue)} + \beta_7\, \textit{ROE}\,) \qquad (1)$$

Description of the independent (X) and control variables (Controls) picked for the model will be provided in the following section. Regarding the format of the data and the estimation method, we will use the probit model and the maximum likelihood estimation procedure. The underlying functional form of the probit regression equation implies the cumulative distribution function of the standard normal distribution, which is:

$$\mathbb{F}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt \qquad (2)$$

According to our hypotheses, the independent variables (determinants) of the probability of change of the credit rating due to the acquisition are in fact the deal-related variables, reflecting the different characteristics of the proposed M&A deal, namely:

Relative deal size – ratio of the deal value to the market capitalization of the acquiring firm 4 weeks prior to the acquisition announcement (*Aktas et al., 2021*).

Payment method – binary variable that takes the value of 0 for deals where the method of payment is 100% cash, 1 otherwise (*Chang et al., 2018*).

Cross-border – binary variable that takes the value of 1 for acquisitions of non-Russian target firms, 0 otherwise (*Chirkova, Chuvstvina, 2013*).

In order for the model to be correctly specified and statistically significant, we also control for other various and important factors, following the previous studies:

Δ*Key rate* – difference between the key rate of the Bank of Russia on the day of the acquisition announcement and 180 days prior to the acquisition announcement (*Rogova, Novichkov, 2020; Xin et al., 2015*). A positive value of this variable ($\Delta > 0$) indicates a tightening of monetary policy, as the key rate was increased since the last 180 days, while a negative value ($\Delta < 0$), on the contrary, indicates a loosening of monetary policy, since the key rate was decreased in the last 180 days. This variable controls for monetary policy conditions and is a macroeconomic-level type of control variables.

Public Target – binary variable that takes the value of 1 if the target is a public company, 0 otherwise (*Kiesel, Riehl, 2024*). This variable controls for complexity in assigning a credit rating to a combined firm after the acquisition and relates to a target-level type of control variables.

ln (Revenue) – natural logarithm of acquirer's sales in the year preceding the acquisition announcement (*Blomkvist et al., 2024*). This variable controls for size of the acquirer and relates to a firm-level type of control variables.

ROE – net income to book value of equity of the acquirer in the year preceding the acquisition announcement (*Chen, 2003*). This variable controls for profitability of the acquirer, taking into account its leverage which is a critical factor in credit rating assignment process, and relates to a firm-level type of control variables.

After the respective model with the independent and control variables described above be estimated, we will be able to receive the predicted probability of the credit rating change of the acquirer as a result of the deal:

$$\mathbb{P}\,(\widehat{\boldsymbol{CR\ Event}} = 1) = \mathbb{F}\,(\hat{\beta}_0 + \hat{\beta}_1\ Relative\ deal\ size + \ldots + \hat{\beta}_7\ ROE\,) \tag{3}$$

The equation above is in fact the *ex-ante* probability, i.e. the *expectations* of investors about the probability of change of the credit rating of the acquirer due to the M&A deal:

$$\mathbb{P}\,(\widehat{\boldsymbol{CR\ Event}} = 1) = \mathbb{E}\,(\boldsymbol{CR\ Event}\,) \tag{4}$$

The concept introduced and articulated above will give us the investors' *expectations*, which we will use as the determining factor of the M&A performance in the subsequent section. To sum up the first step of the research, we will be able to test the hypotheses H1.1–H1.3 using the binary choice model expressed in the equation (1) and also to construct the expected probability of *Credit Rating Event* expressed in the equation (4) used as the M&A performance determinant.

3.2.2. Event study and model for assessing the impact of acquirer's credit rating on M&A deals performance

The second stage is devoted to the main research question of the paper – whether credit ratings influence the M&A performance of acquirers and what is this influence. The foundations laid out in this section will help to reveal this relationship and evaluate its consistency. But before we proceed with the general framework and the model to examine this dependence, we first need to quickly refer to what is the M&A performance and how to measure it.

Following the bulk of authors analyzed in the literature review, the M&A performance of deals in this paper will be measured using the event study (*Aktas et al., 2021, Kiesel, Riehl, 2024, and others*). *Event study analysis* was chosen due to the following reasons and has several benefits:

1) It is one the most popular methods to assess the M&A performance and is a classical method of corporate finance studies proven by decades of research (*MacKinlay, 1997*). It is used both in the developed and the emerging markets (*Skvortsova et al., 2013; Grigorieva, 2020*);
2) It directly measures shareholder value created through stock price performance. Stock prices immediately react and reflect all the news and events related to the company, and

represent the value of the expected *future* cash flows. Therefore, it is a forward-looking measure of value creation (*Brealey, Myers, 1980*);

3) Alternative methods like accounting studies may be subject to manipulation by companies and the use of those methods could be complicated by differences in accounting standards and principles, used in different countries (*Healy, Palepu, 1992*).

However, one of the key drawbacks of this approach is associated with its assumptions. Event study implies the efficiency of the market or, the so-called, efficient market hypothesis (EMH). The questionable moment here lies in the use of this methodology on the emerging markets, e.g. Russia, since these markets, generally, are not considered to be efficient as they are developing. However, there are instances of failure of the EMH even on the developed markets (*Titan, 2015*), so that, rigorously speaking, the proposed approach during *certain periods* might not be used even there. Despite that, given that the main focus of this paper is the credit rating as the determining factor for the M&A performance, the use of an alternative method, i.e. the accounting studies, becomes impossible, since it is roughly indefinable on how to measure the credit rating impact on the accounting statements. The provided arguments conclude that we will stick to the *event study*.

The baseline theoretical foundations for the event study and the cumulative abnormal returns are laid out in (*MacKinlay, 1997*). In this paper we will follow the guidelines stated thereof, while here we will only formulate the key inputs that we implied in our analysis:

The actual (observed) return of stock prices and market index is calculated using the logarithmic returns formula as in (*Skvortsova, Vershinina, 2021*):

$$R_{i,t} = ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \tag{5}$$

The normal (expected) returns are predicted using the market model following the approach used in the research (*Hasan et al., 2021; Blomkvist et al., 2024; Kiesel, Riehl, 2024*):

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t} \tag{6}$$

where $R_{m,t}$ is the return of a market index on day t; α_i – the constant part of the return of company i not explained by the market; β_i – the slope coefficient or the sensitivity of company i to the market; $\varepsilon_{i,t}$ is the random error and $t \in [t_1; t_n]$ is the estimation period.

The estimation period is 250 trading days prior to the event window following the previous studies (*Blomkvist et al., 2024; Kiesel, Riehl, 2024*). This corresponds to a year over the event window, since a standard calendar year (365 days) includes on average 252 trading days. Regarding the benchmark for the market index we will use the MOEX Russia index as the most commonly used benchmark in studies conducted on the Russian market (*Kupatadze, Rogova, 2019; Zagudalova et al., 2024*).

After the OLS regression is run on the equation (6), the estimated coefficients of alpha and beta are obtained. The next step is to calculate the abnormal return ($AR_{i,\tau}$) for each day in the event window and for each M&A deal announcement:

$$AR_{i,\tau} = R_{i,\tau} - \widehat{R}_{i,\tau} = R_{i,\tau} - \hat{\alpha}_i - \hat{\beta}_i R_{m,\tau} \tag{7}$$

where $R_{i,\tau}$ is the actual return and $\tau \in [T_1; T_k]$ is the event window.

We use a 7-day [–3;+3], 11-days [–5;+5], and 21-days [–10;+10] event windows following the previous research carried out in emerging markets (*Reznichenko, 2022; Zagudalova et al., 2024*). Theoretically, the length of the event window depends on such characteristics like information efficiency, ″rumours spread″ and etc. Developing markets, in this respect, are considered to be lower information efficient compared to developed markets, and thus require a longer time to fully encapture the impact of certain event in the stock price. Therefore, we also take a greater length of the event window (up to 21-days) in our research.

The final step is to calculate the cumulative abnormal returns (CAR) of each deal for each of the selected event windows:

$$CAR_i = \sum_{\tau=T_1}^{T_k} AR_{i,\tau} \tag{8}$$

As by the definition, the cumulative abnormal returns measure the stock market reaction to the deal announcement. This is the measure of M&A performance. A positive value of CAR (> 0) indicates that the market prices of shares have grown up as a result of the news about M&A and the market reacted positively to the deal. A negative value of CAR (< 0), on the contrary, implies that the share prices have dropped due to the deal and the deal was negatively perceived by the market.

Finally, after we have defined the measure of M&A performance, we can proceed to the concluding part of this section – identifying the determinants of M&A performance. To assess the

impact of acquirer's credit rating on M&A performance, we will construct the following regression model, where cumulative abnormal return will act as a dependent variable:

$$CAR = \beta_0 + \beta_1\, Credit\ Rating + \beta_2\, \mathbb{E}(CR\ Event) + \beta_3\, Relative\ deal\ size +$$
$$+ \beta_4\, Payment\ method + \beta_5\, Cross\text{-}border + \beta_6\, D/E + \beta_7\, P/E + \beta_8\, ROA + \varepsilon \qquad (9)$$

The model is estimated using the Ordinary Least Squares (OLS) method. The type of data is the cross-sectional, since each M&A deal has its own announcement date.

According to our hypotheses, we introduce the following independent variables encountered in the work of (*Kiesel, Riehl, 2024*), but constructed and adopted with certain improvements to the latter, elaborated here on:

Credit Rating – acquirer's credit rating on the day of the deal announcement measured on a 19-step numerical scale (*Kiesel, Riehl, 2024; Hasan et al., 2021*). See the Appendix 2 for the rating-to-number scale conversion table (*Exhibit 2*). In this paper we used national scale credit ratings assigned by different rating agencies (international CRAs: S&P, Fitch, Moody's; Russian national CRAs: ACRA, RAEX) converted to a common base rating scale. First of all, for each observation we collected all of the available at the time of the M&A announcement credit ratings assigned by each of the 5 rating agencies listed above. This approach allowed us to maximize the sample, as the use of a single rating agency would have significantly cut the number of observations since a certain company may have no rating from the picked CRA, but has a rating in another CRA. Secondly, if the company did not have an assigned national scale CR, we translated the international scale CR into a national one using the international and national scale correspondence tables provided by the rating agencies. An example of such correspondence table is presented in Appendix 2, Exhibit 3 and in the reference list (*S&P, 2017; S&P, 2018*). We also took into account that due to the sovereign CR revisions over the period (country risk changes), the translation scales also changed, so we used the relevant correspondence table for each year. Thirdly, we aggregated the obtained CRs from different rating agencies to a single base scale. In most cases, CRs are consistent and have a high degree of correlation (*Chang et al., 2018*), but sometimes there are some discrepancies in CRs. In case of such discrepancies, we performed the 4th step (*optional*): *(Stage 1)* If assigned ratings from 2 or more CRAs matched, the base scale used that rating level. *(Stage 2)* If the ratings from different CRAs were different, the most recently revised rating was used in the base scale (if it was revised 6 months prior to the deal). *(Stage 3)* If the ratings did not match and one of the ratings had not been revised in the 6 months prior to the deal, the most conservative rating was taken.

There are several papers that used CRs from different rating agencies in their research (*Chen, 2003; Lee et al., 2022*) and developed a methodology to align CRs from different CRAs to a base scale (*Dyachkova, 2018*). It is also a common practice to use Moody's (S&P) rating when there is no S&P (Moody's) rating, assuming they are substituted (*Yu F., 2005; Butler, 2008*).

$\mathbb{E}$*(CR Event)* – binary variable that takes the value of 1 if investors expect a change of the credit rating of the acquirer as a result of the deal, 0 otherwise (*Kiesel, Riehl, 2024*). The variable reflects the investors' *expectations* about negative rating actions and is the estimated probability of acquirer's credit rating change due to the M&A deal (see equation (4)). For details on the variable construction refer to the previous section (*2.2.1*).

As in the previous section, we also introduce the control variables in order for the model to be adequate, obtain good statistical properties and control for other significant factors:

Relative deal size – see page 26 for definition. This variable controls for the size of the deal and is a deal-related type of control variables.

Payment method – see page 26 for definition. This variable controls for the use of alternative methods of payment rather than cash (e.g. equity raise, attracting debt), which means a higher debt burden or an equity dilution, and is a deal-related type of control variables.

Cross-border – see page 26 for definition. This variable controls for geographical diversification and is a deal-related type of control variables.

D/E – ratio of the total liabilities of the acquirer in the year preceding the acquisition announcement to the market capitalization of the acquirer 4 weeks prior to the acquisition announcement (*Grishunin, Egorova, 2022*). This variable controls for financial leverage of the acquirer and is a firm-related type of control variables.

P/E – price-to-earnings multiple, ratio of the acquirer's stock price 1 month before the event window to the EPS of the acquirer in the year preceding the acquisition announcement (*Chirkova, Chuvstvina, 2013*). This variable controls for market valuation of the acquirer and is a firm-related type of control variables.

ROA – net income to total assets of the acquirer in the year preceding the acquisition announcement (*Skvortsova, Vershinina, 2021*). This variable controls for profitability of the acquirer and is a firm-related type of control variables.

The model described above will help us to reveal the main question of the paper, concerning the influence of acquirer's credit rating characteristics on M&A performance (both the absolute level of CR and the probability of CR change as a result of the deal). To summarize the methodology used on the second stage of the empirical research, we will be able to test the hypotheses H2 and H3 using the regression model expressed in the equation (6).

4. Probability of credit rating review as a result of a deal and influence of acquirer's credit rating characteristics on M&A performance in the Russian market in the period from 2000 to 2022

This is the concluding part of the paper which represents the data and the empirical results of the research. We first describe the data collection process and the sources for financial and other necessary information. In the second step, the proposed hypotheses are tested and the models from the previous section are evaluated, afterwards the results are discussed.

4.1. Sample selection criteria and description of the data

For further research, a sample of mergers and acquisitions conducted on the Russian market was formed. As for the sources used in this paper, there is no single database or resource in which all the needed information would be accurately stored and collected.

Initially, we used the *Refinitiv Eikon* database which provided us with over 38,5 thousand M&A transactions conducted worldwide. We further ensure that the deals in our sample fit into the following criteria:

1) Transaction is conducted on the Russian market. We define such transactions after (*Chirkova, Chuvstvina, 2013*) as having the acquirer to be the Russian company, so that either the domestic deals (Russian Acquirer – Russian Target), or the out-bound deals (Russian Acquirer – non-Russian Target) fall into our sample. We do not consider the in-bound deals (non-Russian Acquirer – Russian Target).

2) Transaction is announced and completed during the period of 01/01/2000 – 01/02/2022.

3) Acquirer is a publicly traded company with a credit rating at the date of the deal announcement.

4) Size of the deal is over $1 million. We set this restriction in order to eliminate low value transactions, as small transactions are considered as having an insignificant impact on stock performance and can be neglected (*Teti et al., 2017*).

5) Acquirer owns an equity stake of more than 50% + 1 share of Target after the deal (*Skvortsova, Vershinina, 2021*).

6) Acquirer does not belong to financial (banking) sector. Financial industry operates at different regulations and make their financial statements under different accounting rules (*Skvortsova, Vershinina, 2021; Zagudalova et al., 2024*).

After removing the outliers, our sample consisted of 201 M&A deals which yields the final sample for empirical research.

In addition to the *Refinitiv Eikon* database, which provided us with the set of deals and the financial characteristics of Target and Acquirer needed for construction of the control variables (e.g. EPS, Revenue, Net income, Total Assets, Deal value), we also used the *Investing.com* and *mfd.ru* to download the share prices of acquiring company to evaluate *CAR*. The MarketCap of acquirer used in the denominator of some of control variables was calculated based on the share prices used above and the total number of shares outstanding from *Refinitiv*. Regarding the database for the press releases and news announcements of credit rating agencies about acquirers' credit ratings, we used *CBonds* and *finmarket.ru* to construct the binary variable *CR Event*. The key rate for Δ*Key rate* variable was downloaded from the official website of the Bank of Russia *cbr.ru*.

An additional comment has to be made regarding the currency of returns calculated for *CAR* estimation. Since some of the acquirers which have an operating business in Russia (i.e. they are Russian companies according to the definition of (*Chirkova, Chuvstvina, 2013*) used above) are listed on international stock exchanges (e.g. NYSE, LSE), we used the respective exchange rates ($, €, £) to recalculate the returns in order to correctly compare currency-denominated returns in one basis – the Russian ruble. The exchange rates and the values of Moscow stock exchange index (MOEX Russia) were also taken from the *Investing.com*.

Finally, after we have described the data preparation procedures, we can proceed to the summary of the data. Descriptive statistics of the variables are presented in the Appendix 2, Exhibit 4. The average value of cumulative abnormal returns across the whole sample of 201 deals (CAAR) is not higher than 0.5% by module for each of the three selected event windows, which indicates an overall lack of somewhat significant stock market reaction to an average deal in the sample. Precisely, the stock market reaction is positive (more closely to zero) in case of [–3;+3] and [–10;+10] event windows, and is negative for [–5;+5] event window. This result correlates with the general results obtained in the previous studies of M&A, that indicate an overall absence of reaction to the announcements of M&A deals for acquiring firms' shareholders (*Chirkova, Chuvstvina, 2013; Bertrand, Betschinger, 2012*).

Moving on to the descriptive statistics of control variables and deals' characteristics, the average deal size in our sample is $595 million, which corresponds to the average level of *Relative deal size* of 7.87% (ca. 13 times lower than the average Acquirer Market cap). Out of 201 deals

29 were cross-border (14%) and 57 involved some form of payment other than (or in addition to) cash (28%). There were 24 deals where the Target company was a publicly traded enterprise (12%). Regarding the macroeconomic conditions (Δ*Key rate*), there were 50 deals which saw a tightening of monetary policy on the horizon of 6 months prior to the deal announcement, 106 times the key rate was decreased in the period of 6 months before the deal, and in 45 cases there were neither tightening, nor loosening of monetary policy (i.e. the key rate was left unchanged in the examined period). The average value of changes of key rate on the whole sample of deals equals -0.20%.

The acquirers' financial, profitability and market valuation characteristics are the following. The logarithm of revenue equals 12.25 (mean value). The financial performance in terms of returns on assets (ROA) and equity (ROE) are 9.17% and 26.9%, respectively, for an average acquirer. The financial leverage measured by the book value of liabilities to market cap (D/E) is 0.84x, on average. And the market valuation based on the P/E multiple of acquiring company is 11.0x.

An important set of variables are the credit rating-related measures, where we will discuss the data in a little more details. The median value of *Credit Rating* in our sample is 17, or 'AA' grade (the mean value in this case is meaningless, since CR is a certain positive integer number). The lowest CR in our sample is 8 ('BB'), the highest – 19 ('AAA'). The overall distribution of CRs in our sample looks like the following:

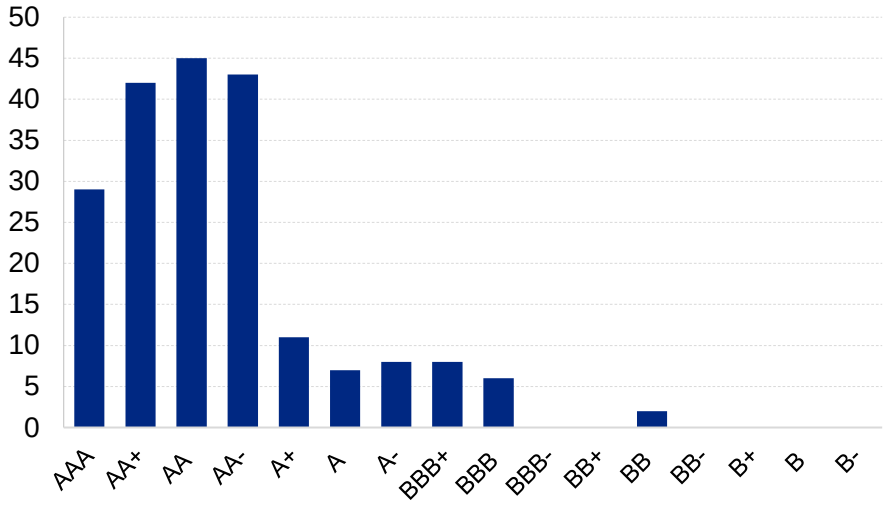

Fig. 1. Distribution of sample by credit quality

Source: authors' calculations

The variable resembles the normal distribution though has a positive skewness (i.e. has a fatter and longer tail on the right side). As can be seen from the graph, almost 96% of our sample

are companies of investment grade ratings (‘BBB+’ and above). This signifies the relative high credit quality of our acquirers and means that the lower-tier companies are generally left out of the sample. So the proposed M&A deals under inspection are also, presumably, of a high investment attractiveness.

Finally, we will refer to the *CR Event* variable, which was, as we mentioned before, constructed based on the review of credit rating agencies' press releases, each of which was manually readthrough by the authors to detect and identify the justifications of credit rating agencies' rating actions related to the deal. Our analysis yields the following data sample:

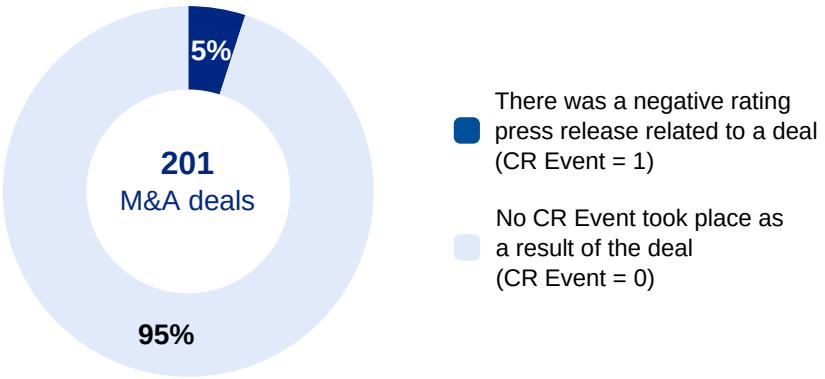

Fig. 2. Negative rating actions after the deal announcement

Source: authors' calculations

As can be seen, only a tiny fraction of deals (5%, or 10 deals) resulted in a negative credit rating agency press release thereafter. However, if they will be matched with the significant negative stock market reaction to the deal announcement, we will reach the desired result regarding the negative influence of probability of acquirer’s credit rating downgrade on M&A performance. In addition, this result corresponds to the result of (*Kiesel, Riehl, 2024*), where the authors have showed that only 17% of acquiring firms in the US market received a downgrade of a credit rating in the first year following the acquisition announcement, indicating a relatively small sample of deals that resulted in a negative rating action, as in our own sample.

Additional and the final step in data analysis will provide clarity on the composition of acquirers in our data. The most frequent industries of acquirers in our sample are TMT (29%), Oil & Gas (27%), Metals & Mining (14%) (*see Exhibit 5*). The following are the most common acquirers in our sample: MTS had conducted 28 deals, Lukoil – 13 deals, Rosneft – 12 deals, VimpelCom (Beeline) – 11 deals (*see Exhibit 6*). Thus, the top-4 acquirers account for over 32% of deals and belong to the TMT or the Oil & Gas sectors.

Before we proceed to the discussion and presentation of the results, we have to confirm that the proposed models will adequately work on the obtained sample and can be used to interpret the results and make conclusions.

A series of tests were carried out to verify the statistical properties of the models. The probit model's maximum likelihood (ML) estimators are consistent if the likelihood function is correctly specified. The most common deviations from that, which arise in the binary choice models, are the heteroskedasticity and non-normality of the estimated residuals of the model.

The White test was performed to test for the presence of heteroskedasticity in the data. The null hypothesis of homoskedasticity was not rejected on all reasonable levels of significance, indicating a homogeneous behaviour of the model's residuals (i.e. constant variance) and thus the uniformity of the data. The residuals were also tested for normality using the Skewness-Kurtosis test similar to the Jarque–Bera test for normality. The results showed that there is no evidence to assume any other distribution other than normal, which means that the model's errors behave like a normal distribution. The probit model thus obtains good statistical qualities and its results can be safely interpreted.

In addition to the probit model, the standard regression model was also tested for conditions necessary to perform the regression analysis and OLS estimation. The heteroskedasticity was rejected based on the same White test. The Breusch–Godfrey test did not confirm the presence of autocorrelation between the errors. The Jarque–Bera test showed the normality of residuals.

Finally, both models were also tested for multicollinearity using the correlation matrix of paired correlations between the models' explanatory variables. The analysis shows that there are no significant correlations between the models' variables higher than 0.286 and 0.537 for the probit and the OLS regressions, respectively (*refer to Ex. 7 and Ex. 8 in Appendix 2*). This means that the models' exogenous variables are uncorrelated to each other and the problem of multicollinearity does not arise. Additionally, the absence of multicollinearity for OLS model was tested using the variance inflation factor (VIF) test, which showed the highest factor of 1.72 for the $\mathbb{E}$*(CR Event)*, which is well below the 2.5 level (*see Exhibit 9*). This confirms there is no multicollinearity.

Concerning the endogeneity problem, we follow the argument of (*Kiesel, Riehl, 2024*), who states that we directly connect the trigger event (announcement of the acquisition) with the CRA response (placing the rating under review), eliminating potential endogeneity concerns as the review process follows a chorological order.

4.2. Hypotheses testing and discussion of the results

The first step in our empirical research is to assess the probability of acquirer's credit rating change due to the M&A announcement and to evaluate the impact of M&A deals characteristics on the probability of such change.

Along with the model presented in the equation (1), we have estimated two more models with the addition of leverage and the credit rating, indicated by the index numbers (2) and (3) in the following table, respectively. The Table 1 reports the results of estimation of the probit models using the maximum likelihood procedure.

Tab. 1. Results of the probit models estimation (*dependent variable – CR Event*)

Variables (N = 201)	(1) ℙ (Y=1)	(2) ℙ (Y=1)	(3) ℙ (Y=1)
Relative deal size	0.678	0.714	0.679
	(0.578)	(0.593)	(0.593)
Payment method	1.464***	1.435***	1.465**
	(0.561)	(0.553)	(0.572)
Cross-border	1.101**	1.057**	1.102**
	(0.494)	(0.501)	(0.509)
ΔKey rate	-0.145	-0.169	-0.145
	(0.185)	(0.194)	(0.187)
Public Target	1.078**	1.028**	1.078**
	(0.508)	(0.517)	(0.510)
ln (Revenue)	-0.289	-0.267	-0.288
	(0.197)	(0.198)	(0.210)
ROE	0.470	0.471	0.471
	(0.625)	(0.654)	(0.638)
D/E		-0.0992	
		(0.233)	
Credit Rating			-0.00130
			(0.118)
Constant	-0.0929	-0.252	-0.0835
	(2.184)	(2.175)	(2.348)
R-squared (pseudo)	0.4320	0.4346	0.4320
LR-test	0.0000	0.0000	0.0000
Observations	201	201	201

standard errors in parentheses *Source: authors' calculations*

*** *p<0.01*, ** *p<0.05*, * *p<0.1*

All of the models are significant at the level of 1% based on the LR-test. However, adding the acquirer's leverage and its credit rating to the (1) model did not provide significant improvement, since the estimated coefficients of the respective variables appeared to be statistically insignificant. This result is contradictory to the result of (*Chen, 2003*) and (*Kang, 2022*), who showed a significant negative influence of leverage on the acquirer's CR downgrade, but is consistent with the results of (*Kiesel, Riehl, 2024*) and (*Chang et al., 2018*), who showed that leverage does not have a significant impact on subsequent acquirer's CR level. Regarding the results of adding the credit rating to the equation, we also received a contradictory result to the one of (*Kiesel, Riehl, 2024*), who showed a significant positive influence of CR on the probability of acquirer's CR downgrade.

To choose between the models, we have performed Akaike's information criterion (AIC) and Bayesian information criterion (BIC) analysis, which showed us that indeed the (1) model is the best, since adding the leverage and credit rating did not significantly improve the explanatory power of the model. So, based on the information criteria, we pick the (1) model for further analysis, as it has the lowest values of AIC/BIC (*see Exhibit 10*).

To analyse and interpret the results of the binary choice models, where the dependent variable is a dummy, we need to calculate the marginal effects (i.e. partial effects), $\partial P(y = 1 \mid x)/\partial x_{i'}$, since the direct values of coefficients are not subject to interpretation. The Table 2 reports the main results of the probit model estimation (for the output from STATA, *see Exhibits 11-12*).

Tab. 2. Results of the probit model (1) estimation

	$\mathbb{P}$ (CR Event=1)	
Variables (N = 201)	Coefficient	$\partial y/\partial x_{i'}$
Relative deal size	0.678	0.040
	(0.578)	(0.033)
Payment method	1.464***	0.086***
	(0.561)	(0.032)
Cross-border	1.101**	0.065**
	(0.494)	(0.028)
ΔKey rate	-0.145	-0.009
	(0.185)	(0.011)
Public Target	1.078**	0.063**
	(0.508)	(0.029)
ln (Revenue)	-0.289	-0.017
	(0.197)	(0.011)
ROE	0.470	0.028
	(0.625)	(0.037)

standard errors in parentheses *Source: authors' calculations*

*** *p<0.01*, ** *p<0.05*, * *p<0.1*

As can be seen from the table above, we reject the H1.1 hypothesis as the *Relative deal size* variable is not significant at each of the significance levels. This result is inconsistent to the results of (*Kiesel, Riehl, 2024; Furfine, Rosen, 2011*), who showed a significant positive influence of relative size on the probability of credit rating downgrade in the US market. This may be partially explained by the following intuition. As large deals usually require debt financing (*Ryan, 2013*), the large deals conducted on the Russian market *presumably* are more often financed with own sources, rather than debt. Therefore, the large size of the deal does not affect the CR position of the acquirer. However, there is no direct evidence to that, since the authors did not perform the statistical analysis of sources of funding of deals on the Russian market. The overall conclusion is that the size of the deal does not have an impact on CR review in the Russian market.

The hypothesis concerning the deal's payment method H1.2, is also rejected since the *Payment method* is significant at 1% level, but has a positive sign, which means that paying in stocks on the Russian market has a significant positive influence on the probability of CR downgrade. This refutes with our expectations and contradicts (*Chang et al., 2018; Furfine, Rosen, 2011*), who showed a significant negative influence of this variable on the US market. Such interesting results on the Russian market may be subject to the following explanation. Paying in stocks is a signal to the market that the acquirer's equity is overvalued, and its intrinsic (true) value is less than the currently observed value on the market. It means that there is an information asymmetry in the market and, therefore, it draws the attention of CRAs to the potential revision of a CR, since it is in their work to eliminate the information asymmetry and act as an intermediaries between the issuers and investors. Consequently, they perform a CR review which results in a negative implication on the acquirer's CR. Thus, we find a significant positive influence of stock payment method on the probability of acquirer's credit rating downgrade on the Russian market. This influence is significant at 1% level and has a marginal effect of 8.6 percentage points increase to the probability of a CR downgrade.

Accordingly, the hypothesis H1.3 concerning the positive influence of cross-border nature of deals on acquirer's credit rating downgrade is not rejected at the 5% significance level. This does not correspond to the result of (*Kiesel, Riehl, 2024*), who showed an insignificant influence of this variable on the US market, but coincides with our expectations. We consider that the announcement of a cross-border deal on Russian market is a significant and new important factor for the rating process, which was not considered by the rating agency prior to the public announcement of this information. Thus, a new factor comes into play and CRAs re-perform their assessment of the

company in the face of new risks. Based on our findings, a cross-border deal conducted by a Russian acquirer is a significant positive factor contributing to the probability of a CR downgrade. This influence is significant at 5% level and is estimated as 6.5 percentage points increase to the probability of a CR downgrade, based on the marginal effects.

In addition to the variables of our interest, we observe a significant positive influence of a *Public Target* acquisition on the probability of acquirer's credit rating downgrade. This influence is significant at 5% level and is estimated as 6.3 percentage points increase. This contradicts the results of previous studies (*Kiesel, Riehl, 2024; Chang et al., 2018; Furfine, Rosen, 2011*), where an insignificant relationship was found. The other controlling variables are insignificant and do not fall to interpretation.

Based on the results of estimation of the probit model (1) from the **Tab. 2** above and using the equation (3) from the previous section, we will construct the binary variable, $\mathbb{E}$ *(CR Event)*, which will denote the investors' expectations regarding the probability of change of acquirer's credit rating.

In order to do so, we constructed a ROC curve of the predicted probabilities based on the model's regression results. ROC curve indicates all of the possible combinations of predicted outcomes by varying the *sensitivity* and *specificity* rates of the model. So, the higher the ROC curve, the better is the predictive power of the model. Based on our analysis, the estimated model yields the following prediction results (for the STATA output, *see Exhibit 13*):

- Area under curve (AUC) is equal to the 0.9555;
- Sensitivity (true positive rate) is equal to the 0.90;
- Specificity (true negative rate) is equal to the 0.93.

This in generally is considered to be an outstanding result for the predictive power of such kind of models. AUC > 0.9 indicates that the model can effectively distinguish between 0's and 1's. We further construct the table of frequency counts in order to illustrate the result (*see Exhibit 14*):

		$\mathbb{P}$ ($\hat{Y}$=1)		
		0	1	Total
$\mathbb{P}$ (Y=1)	0	177	14	**191**
	1	1	9	**10**
	Total	**178**	**23**	

So, based on the table above we can see that indeed our model predicts the 9 out of 10 *Credit Rating Events*, so the sensitivity (share of correctly identified 1's) is 90%. However, there are also 14 cases when the model predicts 1's when the true value is 0, i.e. investors predict that the M&A deal will cause a negative press release, but it doesn't. This is consistent with the fact that we have identified the investors' expectations about the event, not the real *ex-post* variable. In such case, we observe the behaviour of the real market which possesses the information available to investors at the time of the deal announcement, and thus it falls into the timeframe of the event window.

Finally, we approach the ultimate (second) step of our empirical research – determining the influence of acquirer's CR characteristics on M&A performance. The variable calculated in the previous step along with the absolute level of acquirer's CR will be used in the following regression. The Table 3 reports the main results of the regression models estimation (for the output from STATA, *refer to Exhibits 15-17*).

Tab. 3. Results of regression models estimation (*dependent variable – CAR*)

Variables (N=201)	(1) CAR(–3,+3)	(2) CAR(–5,+5)	(3) CAR(–10,+10)
Credit Rating	-0.0031* (0.00157)	-0.00299 (0.00195)	-0.0062** (0.00266)
$\mathbb{E}$ (**CR Event**)	-0.0244* (0.0138)	-0.0314* (0.0171)	-0.0423* (0.0233)
Relative deal size	0.0347** (0.0147)	0.0480*** (0.0182)	0.0663*** (0.0248)
Payment method	0.0115 (0.00892)	0.0163 (0.0110)	0.0104 (0.0151)
Cross-border	-0.0257** (0.0106)	-0.0485*** (0.0131)	-0.0402** (0.0180)
D/E	0.0125*** (0.00337)	0.0106** (0.00416)	0.0139** (0.00569)
P/E	-2.89e-05 (0.000173)	1.24e-05 (0.000214)	-2.05e-05 (0.000293)
ROA	0.140*** (0.0482)	0.198*** (0.0596)	0.285*** (0.0814)
Constant	0.0298 (0.0262)	0.0206 (0.0324)	0.0682 (0.0443)
F-test	0.0000	0.0000	0.0000
R-squared	0.1924	0.2011	0.1697

standard errors in parentheses *Source: authors' calculations*

*** $p<0.01$, ** $p<0.05$, * $p<0.1$

All of the models are significant at the level of 1% based on the F-test. We can also see that the regression coefficients are similar on different event windows both in terms of the significance and the behaviour of the discovered relationships (i.e. the sign of coefficients).

Based on the regression results, we do not reject the H2 hypothesis at the 5% and 10% significance levels in models (3) and (1), respectively, with the exception of the model (2), where an insignificant relationship of acquirer's credit rating and M&A performance was found. The significant negative acquirer's credit rating influence on M&A performance was found in models (3) and (1). This result partially agrees with (*Aktas et al., 2021*) where a curvilinear relationship for the US market was observed, and disagrees with the results of (*Kiesel, Riehl, 2024; Blomkvist et al., 2021; Hasan et al., 2021*) who argued that CR level does not have a significant impact on M&A deals announcement returns in developed markets. However, we are the first study to conduct a similar research in the developing market, e.g. the Russian market. Based on our findings, we see that the Russian acquirers indeed follow the *financial constraints* and *free cash flow* arguments, since the credit rating is found to be negatively related to acquirer announcement returns. This influence is significant and becomes more pronounced as the event window widens.

Accordingly, the hypothesis H3 is also not rejected at the 10% significance level for all of the studied event windows. Based on the results of models (1) – (3), the probability of acquirer's credit rating downgrade, measured from the investors' expectations, has a significant negative influence on acquirers' M&A performance on the Russian market. These results are consistent with the results of (*Kiesel, Riehl, 2024*), who also showed a significant negative influence of a probability of acquirer's credit rating downgrade on the deal announcement returns in the US market. However, in addition to (*Kiesel, Riehl, 2024*), we employ an *ex-ante* estimate of this probability to receive the Russian investors' expectations about a certain negative CR event related to the acquisition announcement. Consequently, we find out that Russian investors in fact anticipate the CR downgrade risk and this risk is indeed negatively reflected in the stock returns. This influence is significant and it persists throughout the whole observation period, as both in the immediate and the longer-term stock market reaction on the Russian market.

In addition to the CR characteristics, we also see an important influence of the control variables. *Relative deal size* has a significant positive contribution to the M&A announcement returns during the observed period on the Russian market, which is consistent with some prior studies (*Aktas et al., 2021*), and inconsistent with the others (*Blomkvist et al., 2021*).

The *Cross-border* nature of the deal is found out to be a significant negative factor for the M&A performance of Russian acquirers in the studied period. This result does not correspond to the bulk of previous authors (*Kiesel, Riehl, 2024; Aktas et al., 2021; Blomkvist et al., 2021; Hasan et al., 2021*) who found an absence of relationship between cross-border character of deal and its performance.

The *D/E* (i.e. *leverage*) of the acquirer is a significant positive factor of M&A performance for Russian firms in the observed period. This, in general, agrees with the previous studies conducted on the US market (*Blomkvist et al., 2024; Blomkvist et al., 2021; Hasan et al., 2021*), who also saw a positive influence of leverage on the M&A performance of acquirers.

The *ROA* (i.e. *profitability*) of Russian acquirers is also a positively related variable to the M&A announcement stock returns in the Russian market. Previous studies have found both the positive (*Kang, 2022*), the negative (*Kiesel, Riehl, 2024; Hasan et al., 2021*), and the absence (*Blomkvist et al., 2024; Blomkvist et al., 2021*) of the relationship of this variable to the M&A deals performance.

5. Conclusion

There is an intensive debate in the academic literature on the impact of M&A deals on acquirer's CR downgrade probability and on the influence of acquirer's CR on M&A performance. They are primarily concerned with developed capital markets. During the course of the literature review, it was found that deals characteristic, such as relative deal size, cross border deals and payment method, have contradictory results on acquirer’s credit rating downgrade probability. Prior research has also obtained ambiguous results on the influence of the acquirer's credit ratings *themselves* on the M&A performance and the negative impact of acquirer's credit rating downgrade risk on deal associated abnormal stock return.

In this study we employ a sample of 201 deals, conducted on the Russian market in the period from 2000 to 2022. The empirical research carried out on this sample yields the following results:

- the relative size of the deal does not have a significant impact on the CR review in the Russian market;
- the payment method is a significant positive factor for the probability of acquirer’s credit rating downgrade on the Russian market. This influence is significant at 1% level and has a marginal effect of 8.6 percentage points increase to the probability of a CR downgrade;
- cross-border deals conducted by the Russian acquirers are significantly positively related to the subsequent probability of acquirer’s CR downgrade. This influence is significant at 5% level and is estimated as 6.5 percentage points increase to the probability of a CR downgrade, based on the marginal effects;
- the credit rating level has a significant negative relationship with the acquirer’s M&A deals performance. This influence is significant and becomes more pronounced as the event window widens;
- the Russian investors in fact anticipate the CR downgrade risk and this risk is indeed negatively reflected in the stock returns.

A key limitation of this paper is the non-in-depth analysis of the CR variable construction. The basic credit rating scale is collected from CRs assigned by different rating agencies without quantitative calculations of default risk. The best practice is to calculate default risks for each rating category of each international and national rating agency and compile a base scale and

correspondence tables based on the intersection of these default risks. The second limitation of our study is the lack of developed model to estimate the company's default risk. This would allow us to estimate the influence of M&A deal on the calculated default risk and the impact of calculated default risk on M&A performance. This would provide us with an opportunity to compare the results obtained for credit rating and calculated default risk and verify their robustness.

Despite the fact that there are active discussions in the scientific community on the problem of relationship between CRs and M&A deals, this field is still poorly studied. The opportunities for future research include studying the effect of credit rating on the overall M&A deal process in emerging markets: target selection process, time to deal completion, method of payment, premium deal size and the efficiency of the M&A deal for target, as well as the influence of M&A deal on the probability of target's credit rating downgrade. The obtained results can be compared with previous research findings in developed markets and corresponded with the results of our study.

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Appendix 1

Table 1. Systematization of existing research: Influence of M&A deal characteristics on the probability of the acquirer's CR downgrade

Article	Data			Dependent variable	Deal-related variables						Target-related variable	Acquirer-related variables			
	# of observations, period	Geography	CRA		Relative deal size	Cross-border	Payment method [1]	CAR [2]	Relatedness [3]	Acquisition dummy	Public target	CR	Size [4]	Leverage [5]	Profitability [6]
Kiesel, Riehl (2024)	5 351 domestic and cross-border M&A deals in 1990-2021	USA	S&P	1 if CR after one year is lower than one day prior to the M&A announcement 0 - otherwise	+	0	0	n/a	+	n/a	0	+	0	0	0
Chang et al. (2018)	1 643 M&A deals in the period 1985-2013	USA	Moody's	1 if CR after 3 or 5 years is upgraded compared to one day prior to the M&A announcement 0 if CR is downgraded	-	n/a	+	+	+	n/a	0	n/a	0	0	n/a
Chen (2003)	101 Hi-tech and non-Hi-tech M&A deals in 1996-2001	Taiwan	S&P and Moody's	The CR index: 0 = B or below, 1 = BB, 2 = BBB, 3 = A, 4 = AA, and 5 = AAA	n/a	n/a	n/a	n/a	-	n/a	n/a	n/a	n/a	-	+
Furfine, Rosen (2011)	1 194 M&A deals in the period 1994-2006	USA	Calculated CR based on Expected Default Frequency	ΔRating is the change in the acquirer's CR from the end of the month prior to the merger announcement to the end of the 6 months after the merger completion	+	n/a	-	n/a	0	n/a	0	n/a	n/a	n/a	n/a
Kang (2022)	7 414 M&A deals in the period 1986-2016	USA	S&P	3-year CR change: equals the difference in the CR notches b/w the deal year and subsequent 3 years	n/a	n/a	n/a	n/a	n/a	0	n/a	n/a	+	-	0

"+" – positive significant influence
" - " – negative significant influence
" 0 " – insignificant influence
"n/a" – not analysed in the paper

Source: authors' analysis

[1] Payment method is a dummy variable that takes the value of 1 if the acquisition is financed at least partially with stock, 0 for deals where the method of payment is 100% cash
[2] CAR is the cumulative abnormal return for the acquiring firm's stock as a result of the deal announcement
[3] Relatedness is a dummy variable that takes the value of 1 if both an acquirer and target belong to the same industry, 0 – otherwise
[4] Size is measured as natural logarithm of the acquirer's revenue or total assets
[5] Leverage is measured as total debt divided by equity or total debt divided by total assets
[6] Profitability is identified as ROA or ROE

Table 2. Systematization of existing research: Summary of research results on the influence of M&A deals on acquirer's stock returns in the Russian market

Article	# of observations	Period	Acquirer's CAR	Influence
Zagudalova et al. (2024)	172	2000-2021	$\overline{CAR}$(–1,+1) = +0.32%*** $\overline{CAR}$(–10,+10) = +0.01%*** $\overline{CAR}$(–15,+15) = –0.09%*** $\overline{CAR}$(–20,+20) = –0.13%**	Positive influence for bidder at event windows (–1,+1), (–10, +10) Negative influence for bidder at event windows (–15,+15), (–20, +20)
Reznichenko E. (2022)	53	2010-2019	$\overline{CAR}$(–1,+1) = +0.67% $\overline{CAR}$(–2,+2) = +2.19%** $\overline{CAR}$(–5,+5) = +2.36%** $\overline{CAR}$(–10,+10) = +1.31%	Positive influence for bidder
Skvortsova, Vershinina (2021)	237	2005-2019	$\overline{CAR}$(–5,+5) = –0.30%***	Negative influence for bidder
Kupatadze, Rogova (2019)	205	2008-2018	$\overline{CAR}$(–1,+1) = +0.35% $\overline{CAR}$(–5,+5) = +0.38%** $\overline{CAR}$(–10,+10) = +0.81%	Positive influence for bidder
Chirkova E., Chuvstvina E. (2013)	56	2005-2011	$\overline{CAR}$(–2,+2) = –0.85% $\overline{CAR}$(–14,+2) = –1.66%	Insignificant influence for all event windows. M&A deals do not create or destroy acquirer's shareholders value
Bertrand, Betschinger (2012)	295	1999-2008	$\overline{CAR}$(–1,+1) = +1.1% $\overline{CAR}$(–2,+2) = +1.4%	Insignificant influence for all event windows. M&A deals do not create or destroy acquirer's shareholders value
** mark significance at 10% level* *** mark significance at 5% level* **** mark significance at 1% level*				

Source: authors' analysis

Table 3. Systematization of existing research: Influence of the acquirer's CR characteristics on M&A deal performance

Article	Data			CR characteristics							Deal-related variables			Target-related variable	Acquirer-related variables	
				Assigned CR			Direct and indirect indicators of CR downgrade risk									
	# of observations, period	Geography	CRA	CR level	CR dummy	Initial CR [7]	Pr (CR Downgrade)	Review for downgrade	Split CR [8]	Negative sign	Relative deal size [9]	Cross-border	Cash deal [10]	Public target	Leverage [11]	Profitability [12]
Kiesel, Riehl (2024)	5 351 M&A deals over the period 1991-2021	USA	S&P	0	n/a	n/a	-	0	n/a	0	0	0	+	-	-	-
Aktas et al. (2021)	6 331 M&A deals over the period 1989-2018	USA	S&P	-/+[13]	n/a	n/a	n/a	n/a	n/a	n/a	+	0	+	-	n/a	n/a
Kang (2022)	7 414 M&A deals over the period 1986-2016	USA	S&P	n/a	n/a	n/a	n/a	n/a	n/a	0	n/a	n/a	+	n/a	0	+
Blomkvist et al. (2024)	6 957 M&A deals over the period 1980-2021	USA	S&P	n/a	n/a	+	n/a	n/a	n/a	n/a	0	-	+	-	+	0
Blomkvist et. al. (2021)	10 598 M&A deals over the period 1990-2016	USA	S&P	n/a	n/a	+	n/a	n/a	n/a	n/a	0	0	+	-	+	0
Blomkvist et al. (2021)	2 959 M&A deals over the period 2000-2017	Europe	S&P	0	n/a	n/a	n/a	n/a	n/a	n/a	-	0	0	0	n/a	n/a
Hasan et al. (2021)	4 929 M&A deals over the period 1996-2015	USA	S&P, Moody's	0	n/a	n/a	n/a	n/a	-	n/a	0	0	+	n/a	+	-
Harford, Uysal (2014)	10 880 M&A deals over the period 1990-2011	Not indicated	All available CRAs	n/a	-	n/a	n/a	n/a	n/a	n/a	+	n/a	+	-	n/a	n/a

"+" – positive significant influence
" - " – negative significant influence
" 0 " – insignificant influence
"n/a" – not analysed in the paper

Source: authors' analysis

[7] Initial Rating is a dummy variable that takes the value of 1 for the first year in the sample a company obtains CR, 0 – otherwise
[8] Split CR is a dummy variable that takes the value of 1 if CRs assigned by several CRAs differ, 0 – otherwise
[9] Relative deal size is measured as either deal value divided by an acquirer's market capitalization, or as deal value divided by an acquirer's total assets
[10] Cash deal is a dummy variable that takes the value of 1 if the acquisition refers to 100% cash financed acquisitions, 0 – otherwise
[11] Leverage is measured as total debt divided by total assets
[12] Profitability is identified as ROA or EBITDA divided by total assets
[13] Curvilinear relationship: negative influence of CR on M&A performance for the low rating range, positive influence for the high rating range

Appendix 2

Exhibit 1. *Examples of negative press releases made by rating agencies, concerning the acquirers' credit ratings in connection with a certain M&A deal*

Cbonds

Quick search for bonds, stocks, funds, indices, companies, ISIN

BOND SCREENER | WATCHLIST | BONDS +2 | STOCKS | ETF & FUNDS | INDICES | NEWS AND RES

S&P Puts Pyaterochka, Subsids On Watch Neg On Poss Acquisition

April 12, 2006 Cbonds

Standard & Poor's Ratings Services said today it placed its 'BB-' long-term corporate credit rating on Pyaterochka Holding N.V., the owner of Russia's largest grocery retail network, on CreditWatch with negative implications. At the same time, Standard & Poor's placed its 'BB-' long-term corporate credit and 'ruAA-' Russia national scale ratings on Pyaterochka's operating subsidiary OOO Agrotorg on CreditWatch with negative implications.

Cbonds

Quick search for bonds, stocks, funds, indices, companies, I

BOND SCREENER | WATCHLIST | BONDS +2 | STOCKS | ETF & FUNDS | INDICES | NEWS AN

Moody's has placed all ratings of TMK on review for possible downgrade

March 17, 2008 Cbonds

Approximately USD 300 million in rated debt securities affected

Moscow, March 17, 2008 – Moody's Investors Service has today placed TMK's Ba3 corporate family rating and B1 rating for loan participation notes on review for possible downgrade following the recent announcement of the acquisition from Evraz Group S.A. of the assets and subsidiaries of IPSCO Tubular's business in the USA for approximately USD 1.2 billion, following

Source: CBonds, S&P, Moody's

Exhibit 2. *The Base scale. Rating-to-number conversion table*

Credit Rating	Numerical scale	*continued…*	
		BBB–	10
AAA	19	BB+	9
AA+	18	BB	8
AA	17	BB–	7
AA–	16	B+	6
A+	15	B	5
A	14	B–	4
A–	13	CCC+	3
BBB+	12	CCC	2
BBB	11	CCC–	1

Source: own estimate

Exhibit 3. *Correspondence table between international and national rating scales (Russia) of international rating agencies (2014)*

№	Moody's Investors Service	Moody's Interfax Rating Agency	Standard & Poor's	Standard & Poor's (national scale)	Fitch Ratings	Fitch Ratings (national scale)
1	Baa1	Aaa.ru	BBB+	ruAAA	BBB+	AAA(rus)
2	Baa2	Aaa.ru	BBB	ruAAA	BBB	AAA(rus) / AA+(rus)
3	Baa3	Aaa.ru	BBB-	ruAA+	BBB-	AA+(rus)
4	Ba1	Aaa.ru / Aa1.ru	BB+	ruAA+	BB+	AA(rus)
5	Ba2	Aa2.ru	BB	ruAA	BB	AA-(rus)
6	Ba3	Aa3.ru	BB-	ruAA-	BB-	A+(rus)
7	B1	A1.ru / A2.ru	B+	ruA+ / ruA	B+	A(rus) / A-(rus)
8	B2	A3.ru / Baa1.ru	B	ruA / ruA-/ ruBBB+	B	BBB(rus) / BBB-(rus)
9	B3	Baa2.ru / Baa3.ru	B-	ruBBB / ruBBB-	B-	BB(rus) / BB-(rus)
10	Caa1	Ba1 / Ba2.ru	CCC+	ruBB+ /../ ruBB-	CCC+	B+(rus)
11	Caa2	Ba3.ru /../ B3.ru	CCC	ruB+ /../ ruB-	CCC	B(rus) / B-(rus)
12	Caa3	Caa1.ru /../ Caa3.ru	CCC-	ruB- /../ ruCCC-	CCC-	B-(rus) /../ CCC-(rus)

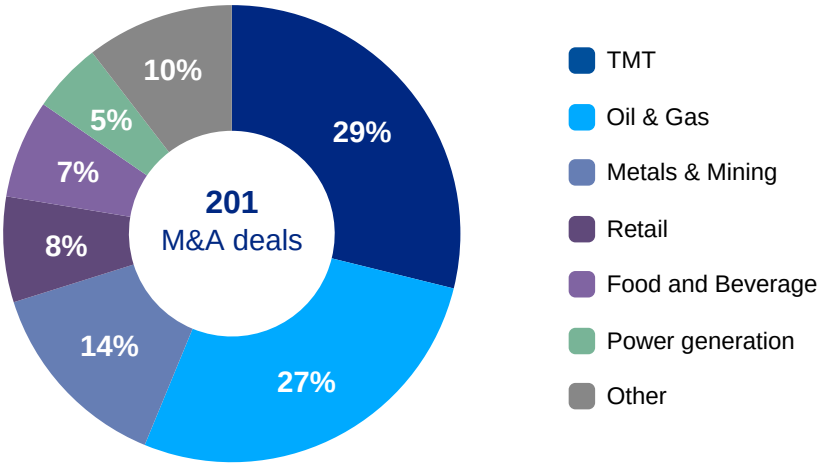
Source: CBonds, Bank for International Settlements data

Exhibit 4. *Descriptive statistics of variables*

Variables	(1) N	(2) Mean	(3) Median	(4) SD	(5) Min	(6) Max
CAR(–10,+10)	201	0.09%	0.18%	8.63%	-20.7%	44.3%
CAR(–5,+5)	201	-0.36%	-0.38%	6.43%	-20.1%	39.3%
CAR(–3,+3)	201	0.15%	-0.06%	5.18%	-13.1%	28.0%
Credit Rating	201	16.44	17	2.181	8	19
CR Event	201	0.050	0	0.218	0	1
$\mathbb{E}$(CR Event)	201	0.114	0	0.319	0	1
Relative deal size	201	7.87%	1.01%	24.5%	0.00%	239.4%
Payment method	201	0.284	0	0.452	0	1
Cross-border	201	0.144	0	0.352	0	1
Δ*Key rate*	201	-0.20%	-0.25%	1.95%	-9.50%	10.25%
Public Target	201	0.119	0	0.325	0	1
ln (Revenue)	201	12.25	12.34	1.628	8.514	15.96
D/E	201	0.84x	0.48x	1.09x	0.00x	9.1x
P/E	201	11.0x	9.08x	19.8x	-100.0x	45.0x
ROA	201	9.17%	8.83%	7.71%	-27.1%	40.9%
ROE	201	26.9%	19.0%	52.5%	-96.5%	429.2%

Source: authors' calculations

Exhibit 5. *Acquirers' industries*



Source: authors' calculations

Exhibit 6. *The most frequent acquiring companies in the sample*

	Acquirer	Number of deals
1.	MTS	28
2.	Lukoil	13
3.	Rosneft	12
4.	VimpelCom	11
5.	Gazprom	8
6.	Cherkizovo	8
7.	Rostelecom	8
8.	Polymetal	8
9.	Golden Telecom	7
10.	X5 Retail	7
11.	Novatek	6
12.	Tatneft	6
13.	Evraz	5
Total		**127**

Source: authors' calculations

Exhibit 7. *Matrix of paired correlations for probit model variables*

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) Relative deal size	1.000						
(2) Payment method	0.130*	1.000					
(3) Cross-border	-0.028	0.087	1.000				
(4) ΔKey rate	-0.028	-0.190***	-0.070	1.000			
(5) Public Target	0.187***	0.143**	0.286***	-0.052	1.000		
(6) ln (Revenue)	-0.185***	-0.163**	0.083	0.116	0.122*	1.000	
(7) ROE	-0.195***	-0.105	-0.021	0.265***	-0.067	0.139**	1.000

*** *p<0.01,* ** *p<0.05,* * *p<0.1*

Source: authors' calculations

Exhibit 8. *Matrix of paired correlations for OLS regression model variables*

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) Credit Rating	1.000							
(2) $\mathbb{E}$(CR Event)	-0.001	1.000						
(3) Relative deal size	-0.133*	0.219***	1.000					
(4) Payment method	-0.027	0.537***	0.130*	1.000				
(5) Cross-border	0.099	0.342***	-0.028	0.087	1.000			
(6) D/E	-0.026	-0.098	0.192***	0.043	-0.183***	1.000		
(7) P/E	0.058	-0.117*	-0.066	-0.074	-0.040	-0.152**	1.000	
(8) ROA	0.156**	-0.019	-0.231***	-0.078	0.240***	-0.318***	0.087	1.000

*** *p<0.01,* ** *p<0.05,* * *p<0.1*

Source: authors' calculations

Exhibit 9. *VIF factors for regressors (OLS regression)*

	VIF	1/VIF
E (CR Event)	1.72	0.581
Payment method	1.44	0.694
Cross-border	1.25	0.803
ROA	1.23	0.815
D/E	1.20	0.833
Relative deal size	1.15	0.866
P/E	1.05	0.954
Credit Rating	1.05	0.956
Mean VIF	1.26	

Source: authors' calculations

Exhibit 10. *AIC and BIC calculation (probit model)*

Akaike's information criterion and Bayesian information criterion

Model	Obs	ll(null)	ll(model)	df	AIC	BIC
model1	201	-39.75421	-22.58215	8	61.1643	87.59074
model2	201	-39.75421	-22.47654	9	62.95307	92.68282
model3	201	-39.75421	-22.58209	9	63.16418	92.89392

Note: N=Obs used in calculating BIC; see [R] BIC note.

Source: authors' calculations

Exhibit 11. *Results of the estimation of probit model from STATA*

```
Iteration 0:   log likelihood = -39.754211
Iteration 1:   log likelihood = -25.680696
Iteration 2:   log likelihood = -22.708639
Iteration 3:   log likelihood = -22.582526
Iteration 4:   log likelihood = -22.582149
Iteration 5:   log likelihood = -22.582149

Probit regression                               Number of obs     =        201
                                                LR chi2(7)        =      34.34
                                                Prob > chi2       =     0.0000
Log likelihood = -22.582149                     Pseudo R2         =     0.4320
```

CRevent	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
Relativedealsize	.6779562	.5779959	1.17	0.241	-.454895	1.810807
Paymentmethod	1.46395	.5613829	2.61	0.009	.3636593	2.56424
Crossborder	1.101012	.4944618	2.23	0.026	.1318845	2.070139
deltaKeyrate	-.1448258	.1850222	-0.78	0.434	-.5074626	.217811
PublicTarget	1.07816	.5083515	2.12	0.034	.0818094	2.074511
lnRevenue	-.2891621	.1966713	-1.47	0.141	-.6746308	.0963066
ROE	.4697785	.6254294	0.75	0.453	-.7560406	1.695598
_cons	-.0929147	2.184221	-0.04	0.966	-4.373909	4.188079

Source: authors' calculations

Exhibit 12. *Marginal effects estimation using STATA*

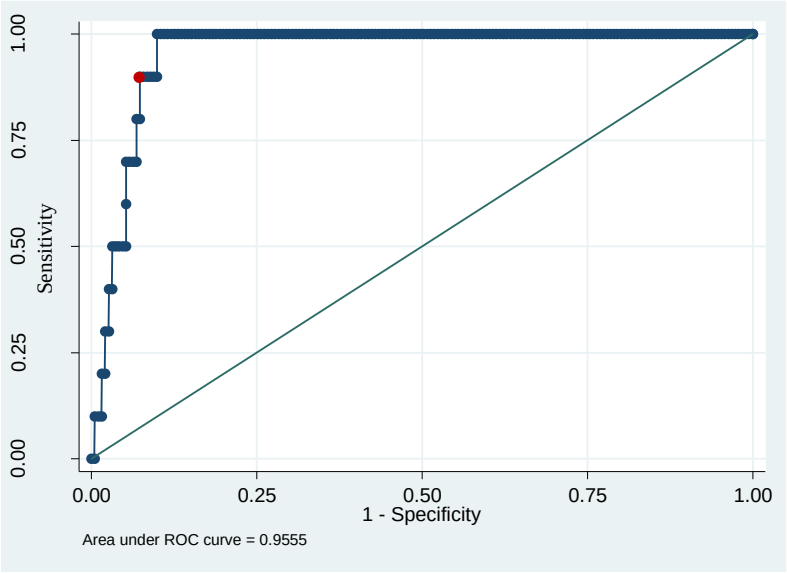
```
Average marginal effects                        Number of obs     =        201
Model VCE    : OIM

Expression   : Pr(CRevent), predict()
dy/dx w.r.t. : Relativedealsize Paymentmethod Crossborder deltaKeyrate PublicTarget
```

	dy/dx	Delta-method Std. Err.	z	P>|z|	[95% Conf.	Interval]
Relativedealsize	.0399083	.0334322	1.19	0.233	-.0256177	.1054343
Paymentmethod	.0861762	.0319895	2.69	0.007	.023478	.1488745
Crossborder	.0648117	.0278886	2.32	0.020	.010151	.1194724
deltaKeyrate	-.0085253	.0108871	-0.78	0.434	-.0298637	.0128132
PublicTarget	.0634665	.0288196	2.20	0.028	.0069812	.1199519
lnRevenue	-.0170217	.0114682	-1.48	0.138	-.039499	.0054556
ROE	.0276538	.0365747	0.76	0.450	-.0440313	.0993389

Source: authors' calculations

Exhibit 13. *ROC (Receiver Operating Characteristics) curve*



Source: authors' calculations

[14] Sensitivity (true positive rate) – proportion of true positive predictions to the sum of true positive and false negative
[15] Specificity (true negative rate) – proportion of true negative predictions to the sum of true negative and false positive

Exhibit 14. *Tabulation output (STATA)*

CR event	ExpCredRatEvent 0	1	Total
0	177	14	191
	92.67	7.33	100.00
	99.44	60.87	95.02
1	1	9	10
	10.00	90.00	100.00
	0.56	39.13	4.98
Total	178	23	201
	88.56	11.44	100.00
	100.00	100.00	100.00

Source: authors' calculations

Exhibit 15. *Results of the estimation of OLS model (dependent variable – CAR*(–3,+3) *)*

```
. reg CAR33 CreditRating ExpCredRatEvent Relativedealsize Paymentmethod Crossborder

      Source |       SS           df       MS      Number of obs   =       201
-------------+----------------------------------   F(8, 192)       =      5.72
       Model |  .103060719         8  .01288259   Prob > F        =    0.0000
    Residual |  .432629937       192  .002253281  R-squared       =    0.1924
-------------+----------------------------------   Adj R-squared   =    0.1587
       Total |  .535690656       200  .002678453  Root MSE        =    .04747

------------------------------------------------------------------------------
       CAR33 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
CreditRating |  -.0030894   .0015736    -1.96   0.051    -.0061932    .0000143
ExpCredRatEvent |  -.0243801   .0137963    -1.77   0.079    -.0515919    .0028317
Relativedealsize |   .0346835      .0147     2.36   0.019     .0056893    .0636777
Paymentmethod |   .0115199   .0089166     1.29   0.198    -.0060672    .0291069
Crossborder |  -.0257019   .0106328    -2.42   0.017     -.046674   -.0047299
          DE |   .0125256   .0033673     3.72   0.000      .005884    .0191672
          PE |  -.0000289   .0001734    -0.17   0.868     -.000371    .0003131
         ROA |    .139512   .0481973     2.89   0.004     .0444478    .2345763
       _cons |   .0297873   .0262345     1.14   0.258    -.0219576    .0815321
------------------------------------------------------------------------------
```

Source: authors' calculations

Exhibit 16. *Results of the estimation of OLS model (dependent variable – CAR*(–5,+5) *)*

```
. reg CAR55 CreditRating ExpCredRatEvent Relativedealsize Paymentmethod Crossborder

      Source |       SS           df       MS      Number of obs   =       201
-------------+----------------------------------   F(8, 192)       =      6.04
       Model |  .166479399         8  .020809925   Prob > F        =    0.0000
    Residual |   .66118886       192  .003443692   R-squared       =    0.2011
-------------+----------------------------------   Adj R-squared   =    0.1679
       Total |  .827668259       200  .004138341   Root MSE        =    .05868

------------------------------------------------------------------------------
       CAR55 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
CreditRating |  -.0029906   .0019453    -1.54   0.126    -.0068276    .0008464
ExpCredRatEvent |  -.0313995   .0170556    -1.84   0.067    -.0650399    .0022409
Relativedealsize |   .0480168   .0181728     2.64   0.009     .0121728    .0838607
Paymentmethod |   .0163163   .0110231     1.48   0.140    -.0054257    .0380582
Crossborder |  -.0484826   .0131447    -3.69   0.000    -.0744092   -.0225561
          DE |   .0105751   .0041628     2.54   0.012     .0023644    .0187857
          PE |   .0000124   .0002144     0.06   0.954    -.0004104    .0004353
         ROA |    .198316   .0595837     3.33   0.001     .0807934    .3158386
       _cons |   .0206052   .0324323     0.64   0.526    -.0433642    .0845745
------------------------------------------------------------------------------
```

Source: authors' calculations

Exhibit 17. *Results of the estimation of OLS model (dependent variable – CAR*(–10,+10) *)*

```
. reg CAR1010 CreditRating ExpCredRatEvent Relativedealsize Paymentmethod Crossborc

      Source |       SS           df       MS      Number of obs   =       201
-------------+----------------------------------   F(8, 192)       =      4.91
       Model |  .252509893         8  .031563737   Prob > F        =    0.0000
    Residual |  1.23532753       192  .006433998   R-squared       =    0.1697
-------------+----------------------------------   Adj R-squared   =    0.1351
       Total |  1.48783742       200  .007439187   Root MSE        =    .08021

------------------------------------------------------------------------------
      CAR1010 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
CreditRating |  -.0062304    .002659    -2.34   0.020    -.0114751   -.0009857
ExpCredRatEvent |  -.0422763   .0233129    -1.81   0.071    -.0882585    .0037059
Relativedealsize |   .0663459   .0248399     2.67   0.008     .0173518    .1153401
Paymentmethod |    .010409   .0150672     0.69   0.490    -.0193094    .0401275
Crossborder |  -.0402306   .0179672    -2.24   0.026     -.075669   -.0047922
          DE |   .0139254     .00569     2.45   0.015     .0027025    .0251484
          PE |  -.0000205    .000293    -0.07   0.944    -.0005985    .0005575
         ROA |   .2849891   .0814433     3.50   0.001     .1243506    .4456277
       _cons |   .0682238   .0443308     1.54   0.125    -.0192142    .1556618
------------------------------------------------------------------------------
```

Source: authors' calculations